

Design and Implementation of a Real-Time Impedance Controller for a 7-DOF Collaborative Robot

*Entwurf und Implementierung eines Echtzeit-Impedanzreglers für einen
7-achsigen kollaborativen Roboter*

Master Thesis

submitted for the degree of Master of Science (M.Sc.)

submitted at

Hochschule München

Faculty of Mechanical, Automotive and Aeronautical Engineering

submitted by

Heykel Khadhraoui

Study programme: Computational Engineering (M.Sc.)

Matriculation number: 41192924

Main supervisor: Prof. Norbert Nitzsche

Second supervisor: Prof. Ruth Otto

Issue date: 1 April 2026

Submission date: 30 September 2026

Declaration of Authorship

I hereby declare that this master's thesis entitled

*“Design and Implementation of a Real-Time Impedance Controller for a 7-DOF
Collaborative Robot”*

was written independently and without assistance from third parties. All sources, references, software tools, and other resources used have been cited accordingly. This work has not previously been submitted as an examination requirement in this or any other form.

Verfassererklärung

Hiermit erkläre ich, dass ich die vorliegende Masterarbeit selbstständig und ohne fremde Hilfe angefertigt habe. Alle verwendeten Quellen, Hilfsmittel, Softwarewerkzeuge und sonstigen Ressourcen wurden entsprechend angegeben. Die Arbeit wurde bisher keiner anderen Prüfungsbehörde in gleicher oder ähnlicher Form vorgelegt.

Munich, 30.09.2026

Heykel Khadhraoui



Abstract

Angular misalignment between a grinding tool and a surface can lead to edge-first contact and increased contact loads. In applications where the surface pose and tool mounting are subject to tolerances, the direction and magnitude of this misalignment are generally unknown before contact. This thesis investigates a Cartesian impedance control approach that uses the contact interaction itself to support the rotational alignment of the tool with a planar surface.

A real-time Cartesian impedance controller was implemented for a seven-degree-of-freedom (DOF) Franka Emika Panda. Translational and rotational compliance are defined relative to the surface geometry. The controller allows the virtual centre of compliance to be shifted away from the tool centre point (TCP). This shift introduces translation-rotation coupling into the Cartesian impedance, so that the commanded normal press can generate an additional moment that supports the required alignment rotation without prescribing an additional rotational trajectory.

The experimental investigation examined the influence of directional Cartesian stiffness and of the tangential position of the virtual compliance centre. The compliance-centre position produced the largest response variation over the tested parameter ranges. An assisting tangential displacement lies perpendicular to the required rotation direction. Reversing the sign of the initial angular deviation requires the displacement to be placed on the opposite side of the TCP, while changing the required tangent-plane rotation direction requires a corresponding change in the displacement direction.

Within the tested conditions, every non-zero tangential displacement that assisted alignment was therefore specific to the initial misalignment direction and sign. The TCP-centred condition introduces no preferred tangential direction and is the direction-independent fixed centre of compliance among the tested positions for the investigated

contact configuration. A displaced centre can instead be selected when the required alignment direction is known and additional rotational alignment authority is needed.

A complementary null-space study showed that projected damping reduced redundant joint motion. Singular-value conditioning was active while the commanded disturbance was applied and strongly suppressed the resulting net redundant displacement under the tested disturbance direction. This response therefore represents disturbance rejection during its application rather than a subsequent return of the redundant configuration.

Kurzfassung

Eine Winkelabweichung zwischen einem Schleifwerkzeug und einer Fläche kann zu Kantenkontakt und erhöhten Kontaktbelastungen führen. Bei Anwendungen, in denen die Lage der Fläche und die Werkzeugaufnahme Toleranzen aufweisen, sind Richtung und Größe dieser Winkelabweichung vor dem Kontakt in der Regel nicht bekannt. Diese Arbeit untersucht einen kartesischen Impedanzregelungsansatz, bei dem die Kontaktinteraktion selbst die rotatorische Ausrichtung des Werkzeugs an einer ebenen Fläche unterstützt.

Für einen 7-achsigen Franka Emika Panda wurde ein kartesischer Echtzeit-Impedanzregler implementiert. Die translatorische und rotatorische Nachgiebigkeit wird relativ zur Flächengeometrie definiert. Das virtuelle Nachgiebigkeitszentrum kann gegenüber dem Werkzeugbezugspunkt (TCP) verschoben werden. Dadurch entsteht eine Kopplung zwischen Translation und Rotation innerhalb der kartesischen Impedanz, sodass die kommandierte normale Anpressbewegung ein zusätzliches Moment erzeugen kann, das die erforderliche Ausrichtungsrotation unterstützt, ohne eine zusätzliche rotatorische Trajektorie vorzugeben.

Experimentell wurden der Einfluss der richtungsabhängigen kartesischen Steifigkeit sowie der tangentialen Lage des virtuellen Nachgiebigkeitszentrums untersucht. Innerhalb der geprüften Parameterbereiche führte die Lage des Nachgiebigkeitszentrums zur größten Variation der Ausrichtungsreaktion. Eine unterstützende tangentielle Verschiebung liegt senkrecht zur erforderlichen Drehrichtung. Wird das Vorzeichen der anfänglichen Winkelabweichung umgekehrt, muss die Verschiebung auf die gegenüberliegende Seite des TCP gelegt werden. Ändert sich die erforderliche Drehrichtung in der Tangentialebene, muss sich entsprechend auch die Richtung der Verschiebung ändern.

Unter den untersuchten Bedingungen war damit jede nicht verschwindende tangentielle Verschiebung, die die Ausrichtung unterstützte, von Richtung und Vorzeichen der anfäng-

lichen Winkelabweichung abhängig. Die TCP-zentrierte Einstellung gibt keine bevorzugte tangential Richtung vor und stellt unter den getesteten Lagen das richtungsunabhängige feste Nachgiebigkeitszentrum für die untersuchte Kontaktkonfiguration dar. Ein verschobenes Nachgiebigkeitszentrum kann dagegen gezielt eingesetzt werden, wenn die erforderliche Ausrichtungsrichtung bekannt ist und zusätzliche rotatorische Ausrichtungswirkung benötigt wird.

Eine ergänzende Nullraumuntersuchung zeigte, dass die projizierte Dämpfung redundante Gelenkbewegungen reduzierte. Die Singularwert-Konditionierung war während der kommandierten Störung aktiv und unterdrückte unter der untersuchten Störungsrichtung die daraus resultierende Netto-Verschiebung im Nullraum deutlich. Dieses Verhalten beschreibt somit eine Unterdrückung der Störwirkung während ihrer Einwirkung und keine nachträgliche Rückführung der redundanten Konfiguration.

Contents

Declaration of Authorship	III
Abstract	V
Kurzfassung	VII
List of Figures	XIV
List of Tables	XVI
List of Abbreviations	XVIII
List of Symbols	XIX
1 Introduction	1
1.1 Motivation	1
1.2 Related Work	2
1.3 Problem Statement	3
1.4 Scope and Contributions	4
1.5 Thesis Structure	5
2 Theoretical Background	6
2.1 Reference Frames	6
2.2 Rigid-Body Pose and Pose Error	8
2.2.1 Cartesian Pose Error	8

2.3	Kinematics of a 7-DOF Manipulator	10
2.4	Cartesian Impedance Control	11
2.4.1	Impedance Wrench	11
2.4.2	Joint-Torque Control Law	12
2.4.3	Coriolis and Gravity Compensation in libfranka	13
2.5	Surface Frame	14
2.6	Stiffness and Damping Representation	15
2.7	Centre of Compliance	16
2.7.1	Commanded Moment	18
2.7.2	Surface-Frame and Tangential Compliance-Centre Displacement	19
2.8	Null-Space Torque	22
2.8.1	SVD and Null-Space Direction	23
2.8.2	Pseudoinverse and Projectors	23
2.8.3	Projected Damping and Conditioning	24
2.8.4	Complete Null-Space Torque	25
2.9	Complete Joint-Torque Command	25
3	Controller Design and Real-Time Implementation	26
3.1	Controller Architecture and Real-Time Control Cycle	26
3.2	Surface-Contact Sequence	28
3.2.1	Surface and Tool Reference	30
3.2.2	Tool Orientation	31
3.2.3	Surface Approach, Clearance and Tool-Point Selection	32
3.2.4	Contact Set-Up	35
3.2.5	Grinding	39
3.3	Cartesian Pose Hold	39
3.4	Robot-Side Safety and Command Handling	41
3.5	Real-Time Data Recording	41

4	Experimental Methodology	42
4.1	Experimental System	42
4.1.1	Robot, Tool, and Surface	42
4.1.2	Real-Time Computer and Software	44
4.1.3	Operating and Safety Configuration	44
4.2	Surface Reference and Tool Calibration	44
4.2.1	Surface-Reference Construction	45
4.2.2	Tool Normal Calibration	46
4.2.3	Calibrated Experimental Geometry	46
4.3	Common Contact-Test Configuration and Procedure	46
4.3.1	Common Cartesian Gains and Damping	47
4.3.2	Contact Sequence and Set-Up Motion	47
4.3.3	Repetitions and Data Recording	50
4.4	Main Surface-Contact Experimental Matrix	50
4.5	Evaluation Quantities	51
4.5.1	Signed Set-Up Response	52
4.5.2	Supporting Geometric Checks	53
4.5.3	Commanded Wrench Quantities	53
4.6	Secondary Null-Space Pose-Hold Experiment	54
4.6.1	Experimental Configuration	54
4.6.2	Commanded Disturbance	54
4.6.3	Evaluation Quantities	56
5	Results and Discussion	59
5.1	Main Surface-Contact Results	60
5.1.1	Baseline Contact Response with the CoC at the TCP: Case A . . .	60
5.1.2	Effect of Cartesian Stiffness: Cases B and C	61
5.1.3	Effect of Tangential CoC Position: Case D	62

5.2	Null-Space Pose-Hold Results	65
5.2.1	Projected Null-Space Damping	67
5.2.2	Singular-Value Conditioning	67
5.2.3	Interpretation and Experimental Bounds	68
6	Conclusion and Future Work	70
6.1	Conclusion	70
6.2	Limitations	71
6.2.1	Experimental Range	71
6.2.2	Measurement and Tool-Mount Uncertainty	71
6.2.3	Null-Space Configuration and Real-Time Operation	71
6.3	Future Work	72
	Bibliography	73
A	Controller Source Listings	77
A.1	Centre of Compliance and the Offset Adjoint	77
A.2	Core Control Callback	78
B	Recorded Experimental Data	80
B.1	Control Phases	80
B.2	Recorded Signals	80
C	Controller Parameters	83
C.1	Surface and Tool Geometry	83
C.1.1	Calibration Relations	84
C.1.2	Initial Joint Configuration	85
C.2	Phase Sequence and Impedance	85
C.3	Inertia-Scaled Damping and Centre of Compliance	86
C.4	Null-Space and Operating Parameters	87

D Supporting Contact Checks and Results	88
D.1 Numerical Values for Main Cases A–D	88
D.2 Orientation-Offset Magnitude	90
D.3 Compliance-Centre Definition Frame	91
D.4 Tool-Axis Displacement	92
D.5 Intermediate Tangent-Plane Directions	93
D.6 Conditions Classified as Tilted by TCP Height	95
D.7 Pose-Based Alignment Consistency Check	96

List of Figures

2.1	Frames used in the controller and the end-effector pose error.	7
2.2	Compliance-centre displacement and commanded TCP moment	19
2.3	Normal-force contribution for opposite tangential displacements	21
3.1	Torque contributions combined in one control cycle.	27
3.2	Surface-contact phase sequence.	29
3.3	Calibrated tool face and its four corners.	31
3.4	Principal selected tool points for the rectangular tool face.	34
3.5	Set-up reference generation for the selected tool point.	37
3.6	Cartesian pose hold with four selectable null-space configurations.	40
4.1	Experimental system and surface-frame contact geometry.	43
4.2	Surface-reference construction and tool-normal calibration.	45
5.1	Measured set-up rotation with the CoC at the TCP.	60
5.2	Measured set-up rotation by rotational stiffness.	61
5.3	Measured set-up rotation by cross-axis translational stiffness.	62
5.4	Set-up rotation by tangential CoC position (mean $\pm$ SD).	63
5.5	Measured set-up rotation, commanded normal force, and commanded TCP moment for three tangential CoC positions.	64
5.6	Null-space pose-hold response.	66
D.1	Set-up rotation by commanded orientation-offset magnitude.	90

D.2	Set-up rotation for tool- and surface-fixed displacement definitions.	91
D.3	Tangential and tool-axis centre displacement at the same command.	92
D.4	Selected tangential displacement for each commanded rotation direction. . .	94
D.5	Set-up rotation with the direction-selected and fixed displacements.	95
D.6	Pose-based alignment estimate against measured set-up rotation.	96

List of Tables

4.1	Calibrated tool geometry.	46
4.2	Common Cartesian gains and damping ratio by phase.	47
4.3	Common phase-transition and trajectory parameters for the surface- contact experiments.	48
4.4	Main surface-contact cases and parameter variations.	51
4.5	Isotropic Cartesian gains of the pose-hold trials.	54
B.1	Recorded signal groups.	80
C.1	Surface and tool parameters.	84
C.2	Phase and mode gains.	85
C.3	Phase-transition and trajectory parameters.	85
C.4	Damping and compliance-centre parameters.	86
C.5	Null-space and operating parameters.	87
D.1	Case-A baseline response at the TCP, mean $\pm$ SD.	89
D.2	Case-B rotational-stiffness response, mean $\pm$ SD.	89
D.3	Case-C cross-axis translational-stiffness response, mean $\pm$ SD.	89
D.4	Case-D tangential compliance-centre response, mean $\pm$ SD.	90
D.5	Offset-magnitude response at two compliance-centre positions.	90
D.6	Definition-frame response at a common 40 mm magnitude.	91
D.7	Tool-axis displacement response at a $+10^\circ$ command about t_1	92

D.8	Commanded directions and the tangential displacements tested against them.	94
D.9	Direction-rule comparison at $\ r_{c,t}\ = 40$ mm.	94
D.10	Conditions classified as tilted by TCP height.	96

List of Abbreviations

CoC	Centre of Compliance
CSV	Comma-Separated Values
DH	Denavit–Hartenberg
DOF	Degree of Freedom
EE	End-Effector
E-stop	Emergency Stop
FCI	Franka Control Interface
FCU	Franka Control Unit
F/T	Force/Torque
IP	Internet Protocol
LDLT	Lower–Diagonal–Lower-Transpose Factorisation
OS	Operating System
PREEMPT_RT	Real-Time Preemption Patch
ROS	Robot Operating System
SE(3)	Special Euclidean Group in three dimensions
SO(3)	Special Orthogonal Group in three dimensions
SVD	Singular Value Decomposition
TCP	Tool Centre Point

List of Symbols

Frames and Kinematics

$\{0\}$	Robot base frame	[-]
$\{EE\}$	End-effector frame used by the robot model	[-]
$\{d\}$	Desired end-effector and TCP reference frame	[-]
$\{S\}$	Surface frame, with axes t_1 , t_2 and n_s	[-]
n	Number of joints ($n = 7$)	[-]
$q, \dot{q}$	Joint position / velocity vector	[rad], [rad/s]
p	Position / translation vector	[m]
R	Rotation matrix	[-]
T	Homogeneous transformation matrix	[-]
$J(q)$	Geometric Jacobian; linear rows above angular rows	[m/rad], [-]

Pose and Error

p_{EE}, p_{TCP}	End-effector / tool-centre-point position; $p_{TCP} = p_{EE}$ in the implemented model	[m]
R_{EE}, T_{EE}	End-effector orientation / transformation	[-]
$R_{EE,clearance}, p_{TCP,clearance}$	End-effector orientation captured at the clearance transition and held as the set-up orientation reference / TCP position captured at the same instant	[-], [m]
$R_{set,start}, R_{set,end}$	Measured end-effector orientations at the beginning / end of set-up	[-]
R_{set}	Current-to-reference relative rotation over set-up, $R_{set,start} R_{set,end}^\top$	[-]
u_{set}, ϕ_{set}	Axis / angle of R_{set}	[-], [rad]
p_d, R_d, T_d	Desired position / orientation / transformation	[m], [-]
ΔR	Current-to-desired relative rotation, $R_{EE}^\top R_d$	[-]
ϕ	Orientation-error angle	[rad]
$u, [u]_\times$	Current-EE-frame orientation-error axis / its skew matrix	[-]
e_p	Translational (position) error	[m]
e_R	Base-frame rotational error used by the command	[rad]
$\dot{x}$	Cartesian velocity (twist)	[m/s], [rad/s]

Cartesian Impedance Gains

K_p, D_p	Generic translational stiffness / damping block; a frame or phase index is added when needed	$[\text{N/m}], [\text{N s/m}]$
K_R, D_R	Generic rotational stiffness / damping block; a frame or phase index is added when needed	$[\text{N m/rad}], [\text{N m s/rad}]$
$K_{p,t_1}, K_{p,t_2}, K_{p,n}$	Translational stiffness entries along t_1, t_2 , and n_s	$[\text{N/m}]$
$K_{R,t_1}, K_{R,t_2}, K_{R,n}$	Rotational stiffness entries about t_1, t_2 , and n_s	$[\text{N m/rad}]$
$D_{p,t_1}, D_{p,t_2}, D_{p,n}$	Translational damping entries along t_1, t_2 , and n_s	$[\text{N s/m}]$
$D_{R,t_1}, D_{R,t_2}, D_{R,n}$	Rotational damping entries about t_1, t_2 , and n_s	$[\text{N m s/rad}]$
K_0, D_0	Complete base-frame Cartesian stiffness / damping, assembled block diagonal from the rotated translational and rotational blocks	$[\text{N/m}], [\text{N m/rad}];$ $[\text{N s/m}], [\text{N m s/rad}]$

Dynamics and Torques

F	Cartesian wrench	[N], [N m]
f, m	Complete commanded Cartesian force / moment at the TCP, with $m = m_R + r_c \times f$	[N], [N m]
f_n, f_t	Normal / tangential component of the complete commanded force, $f_n = F_n n_s$ and $f_t = f - f_n$	[N], [N]
K, D	Full Cartesian stiffness / damping matrices	K : [N/m], [N m/rad]; D : [N s/m], [N m s/rad]
τ	Joint torque vector	[N m]
τ_c, τ_g	Coriolis / gravity compensation torque	[N m]
τ_{cart}	Cartesian impedance torque, $J^\top(q)F$	[N m]
τ_{cmd}	Complete commanded joint torque	[N m]

Surface and Contact Geometry

n_s	Calibrated surface-plane normal [-] copied into the controller and used in the evaluation
t_1, t_2	Surface tangent directions (unit) [-]
R_{surface}	Surface-frame orientation $[t_1 \ t_2 \ n_s]$ [-] in the base frame
$n_{\text{Tool,EE}}, n_{\text{Tool,0}}, n_d$	Tool-face normal expressed in the [-] EE frame, fixed by calibration / the same direction expressed in the base frame, $n_{\text{Tool,0}} = R_{\text{EE}} n_{\text{Tool,EE}}$ / de- sired tool-normal direction used by the controller
$\theta_{\text{cmd}}, u_{\text{cmd}}, R_{\text{cmd}}$	Commanded orientation-offset rota- [rad], [-], [-] tion vector, $\theta_{\text{cmd}} = \theta_{t_1} t_1 + \theta_{t_2} t_2$ / its unit rotation axis / the rotation ma- trix it produces through $R(u, \phi)$
p_s	Point on the configured surface [m]
$e_{x_0}, e_{y_0}, e_{z_0}$	Unit vectors along the base-frame [-] axes x_0, y_0 and z_0
p_c	Centre of compliance [m]
r_c	TCP-to-compliance-centre displace- [m] ment, $p_c - p_{\text{TCP}}$, as the point-shift transformation and the experimen- tal plots and tables use it; where no frame index is shown, the base-frame representation is implied

$r_{c,0}, r_{c,S}, r_{c,EE}$	Representations of the same displacement in the base, surface and end-effector frames, with $r_{c,0} = R_{\text{surface}}r_{c,S} = R_{EE}r_{c,EE}$	[m]
$r_{c,t_1}, r_{c,t_2}, r_{c,n}$	Surface-frame components of r_c along t_1, t_2 and n_s	[m]
$r_{\text{Tool},EE}, p_{\text{Tool}}$	Selected tool-point offset from the TCP, expressed in the EE frame / selected tool-point position in the base frame	[m]
r_{Tool}	Selected tool-point offset from the TCP, $p_{\text{Tool}} - p_{\text{TCP}}$	[m]
$p_{\text{Tool},\text{clearance}}, p_{\text{Tool},0}, p_{\text{Tool},d}$	Selected tool point at the clearance transition / its projection onto the calibrated surface / its desired set-up position	[m]
$s_{\text{app}}, v_{\text{app}}, s_{\text{app},\text{max}}$	Commanded approach travel of the TCP along $-n_s$ / its configured rate / its configured maximum	[m], [m/s], [m]
$p_{\text{TCP},\text{start}}, t_{\text{app},0}, t_{\text{app},\text{max}}$	TCP position captured when surface approach begins / that instant / the instant at which s_{app} reaches $s_{\text{app},\text{max}}$	[m], [s], [s]
h_i	Signed clearance of tool-face corner i from the calibrated surface	[m]
h_{Tool}	Minimum clearance of the rectangular tool face from the calibrated surface, $h_{\text{Tool}} = \min_i h_i$	[m]
$h_{\text{clearance}}$	Configured tool clearance at which the transition from approach to set-up occurs	[m]

ε_{sel}	Tool-point selection tolerance	[m]
$s_{\text{set}}, v_{\text{set}}, s_{\text{set,max}}$	Commanded set-up press coordinate of the selected tool point along $-n_s$ / its configured rate / its configured endpoint	[m], [m/s], [m]
$t_{\text{set},0}, t_{\text{set,max}}$	Instant of the clearance transition / instant at which s_{set} reaches $s_{\text{set,max}}$	[s], [s]
m_R	Rotational-impedance contribution to the commanded moment, $K_R e_R -$ $D_R \omega_{\text{EE}}$, present at every lever set- ting	[N m]

Compliance and Alignment

$\text{Ad}(r_c)$	TCP-to-compliance-centre point-shift adjoint	[-]
K_c, D_c	Block-diagonal stiffness / damping at the centre of compliance	K : [N/m], [N m/rad]; D : [N s/m], [N m s/rad]
F_c, F_{TCP}	Cartesian wrench at the centre of compliance / corresponding wrench at the TCP	[N], [N m]
$K_{\text{TCP}}, D_{\text{TCP}}$	Centre-of-compliance gains mapped by congruence to the TCP	K : [N/m], [N m/rad]; D : [N s/m], [N m s/rad]
$\theta_{t_1}, \theta_{t_2}$	Surface-frame components of the commanded tool orientation offset	[rad]
γ_0, γ_S	Signed set-up response vector in base / surface coordinates: the end-of-set-up orientation change relative to the held set-up reference	[rad], [rad]
$\gamma_{t_1}, \gamma_{t_2}, \gamma_n$	Surface-frame components of the signed set-up response; a component reducing a commanded offset carries the same sign as that offset	[rad]
$r_{c,t}$	Tangential part of the compliance-centre displacement, $r_{c,t_1}t_1 + r_{c,t_2}t_2$; its magnitude is written $\ r_{c,t}\ $	[m]
F_n, M_{t_i}	Signed commanded normal force / commanded TCP moment about surface tangent t_i	[N], [N m]

Null-Space and Singular Value Decomposition

J^+	Thresholded-SVD Moore–Penrose pseudoinverse	[rad/m], [-]
N_q, N_τ	Joint-velocity / torque null-space projector	[-]
$\sigma_i, \sigma_{\min}$	Jacobian singular value / σ_6 conditioning indicator. The geometric Jacobian mixes linear and angular rows, so a singular value carries no single physical unit and its numerical value depends on the chosen Jacobian representation and length unit; it is used as a relative indicator under one fixed convention	[-]
v_7	Structural null direction of the seven-joint Jacobian	[-]
$\dot{q}_{\text{null}}$	Projected joint velocity in the numerical null space	[rad/s]
d_{null}	Null-space damping	[N m s/rad]
k_σ	Sigma-conditioning torque magnitude	[N m]
α_{probe}	Null-direction sampling angle	[rad]
$\Delta\sigma_{\min, \text{dist}}$	Experimental change in $\sigma_{\min}$ over the disturbance interval, $\sigma_{\min}(9\text{ s}) - \sigma_{\min}(5\text{ s})$; distinct from the controller’s Δ_σ	[-]
$\tau_d, \tau_\sigma, \tau_{\text{null}}$	Projected damping / sigma / total null-space torque	[N m]

E_N	Cumulative projected null-space motion; integrated projected joint-velocity magnitude during the disturbance interval, converted to degrees for reporting	[°]
v_{ref}	Fixed redundant comparison direction, taken from the condition without null-space torque	[-]
$\Delta q_{\text{null}}, \Delta \eta_{\text{dist}}$	Net projected joint displacement over the disturbance interval / its signed component along v_{ref}	[rad]

Chapter 1

Introduction

1.1 Motivation

Industrial manipulators commonly use position control to track commanded trajectories. In surface-finishing applications, placement tolerances can cause the physical surface pose to differ from its programmed pose. The surface is positioned manually, and the tool is held in a gripper. Their residual angular misalignment is therefore unknown before contact. Rigid position control resists the corresponding corrective motion and can generate high contact loads.

Torque-controlled collaborative robots allow contact compliance to be specified directly and account for a growing share of industrial deployments [12, 24]. Joint torque sensing and a torque-level command interface determine how the robot yields under contact as well as where it moves. In surface finishing, this capability can reduce the loads caused by angular tool–surface misalignment. It can also permit the tool face to rotate towards the surface while maintaining the commanded normal press. The resulting alignment is evaluated relative to the physical surface.

Impedance control specifies this yielding behaviour as a dynamic relation between motion and interaction force [11]. Cartesian impedance control defines the relation for the end-effector pose. Its translational and rotational compliance can therefore be assigned independently along selected Cartesian directions. During edge-first contact, the wrench contains both force and moment components. Gains defined from the contact geometry allow part of this moment to rotate the tool towards the surface. The stiffness and

damping matrices consequently determine both the contact response and the free-space tracking behaviour.

1.2 Related Work

The relevant literature comprises three areas: Cartesian impedance control, real-time implementation on torque-controlled robots, and the placement of the centre of compliance (CoC). The first area is Cartesian impedance control itself. The controller uses standard serial-manipulator kinematics [21, 3, 17]: homogeneous transformations for representing the end-effector pose and the Jacobian for relating joint motion to Cartesian motion. Khatib established the operational-space formulation, in which a Cartesian wrench F is mapped to joint torques τ through $J^\top$ [13]; this mapping is adopted in the controller. Hogan defined the impedance relation between end-effector motion and interaction force [11]. Albu-Schäffer et al. implemented Cartesian impedance control using joint-torque sensing [1], which corresponds to the torque-controlled interface used in this thesis.

The second area concerns the real-time implementation of Cartesian impedance control on torque-controlled robots. Mayr et al. implemented a modular Cartesian impedance controller for the Franka Emika Panda that combines real-time model access, null-space control, signal treatment, and command limiting [15]. The controller developed in this thesis uses the libfranka model interface to obtain the required robot state and model quantities during the real-time control loop. It also includes a selectable null-space torque τ_{null} for influencing the redundant joint motion without changing the commanded Cartesian motion.

Recent work has applied compliant and force-controlled strategies directly to robotic surface finishing. Alt et al. combined perception, planning, and force-controlled execution for robotic surface treatment [2]. Min et al. integrated demonstrated motion and force skills with impedance control on a 7-DOF collaborative robot [16]. Wu et al. used adaptive variable impedance control to regulate the contact force during robotic grinding [27]. These studies address force-controlled surface treatment. The present work instead examines contact-induced angular alignment through directional Cartesian stiffness and a virtual CoC.

The third area concerns the location of the CoC. Fasse and Broenink formulated spatial impedance about a selected point, so that translation and rotation become coupled through the position of this point [5]. This formulation provides the theoretical basis for the compliance-centre shift used in this thesis. The same principle appears mechanically in remote-centre-compliance devices, in which elastic elements are arranged so that the tool behaves as if it rotates about a virtual point located away from the mechanism itself [18, 26]. More recent mechanically compliant and variable-compliance-centre strategies apply this principle to peg-in-hole insertion [25, 10]. In the mechanically compliant mechanisms, elastic bending produces the alignment motion rather than software control alone. In these approaches, the compliance is arranged so that contact forces reduce misalignment instead of increasing it. Most of these studies focus on insertion tasks and define the compliance centre through either the mechanical design or a strategy adapted to the hole geometry. In this thesis, an equivalent effect is created in software by shifting the virtual CoC away from the TCP. The independent effects of this position and the axis-specific stiffness during planar tool alignment are then investigated on a torque-controlled robot. Suomalainen et al. experimentally investigated how compliance-centre placement and compliant-axis selection influence alignment in an assembly task [23]. Friedrich et al. subsequently proposed an active remote CoC that combines passive compliance with active force control [9]. These approaches further illustrate the role of compliance location in contact alignment. The present work examines a software-defined virtual compliance centre for planar tool-surface alignment.

1.3 Problem Statement

The angular relation between the tool and the surface plane is uncertain before contact because the surface placement and tool mounting both carry tolerances. The sign of the initial angle and its tangent-plane rotation direction are therefore unknown when a run starts. Under rigid position control, the contact geometry can generate high moments while the controller resists the corrective motion. In contrast, finite rotational compliance allows contact-induced rotation, whereas a displaced CoC adds a selectable coupling between translation and rotation.

The lever r_c is the displacement from the TCP to the virtual CoC. A tangential lever

enters the commanded wrench through a cross product with the translational force and thereby selects an additional rotational moment. The direction and sign of this moment depend on the lever direction. A displaced centre chosen before contact therefore encodes a preferred alignment direction.

The investigation evaluates a fixed CoC for conditions in which the sign and tangent-plane direction of the initial angular misalignment are unknown. In this thesis, a direction-independent fixed centre denotes a centre position that can be selected without this prior information. The assessment is limited to the investigated contact configuration, parameter range, and tested centre positions. The term describes direction independence of the selected reference and carries no claim of global optimality.

Contact-induced alignment forms the set-up stage of the intended surface-contact sequence. The tool approaches the surface with an angular offset, establishes contact, and rotates under the compliant interaction. Sustained tangential motion would follow after alignment. However, the reported measurements cover the contact set-up stage and end at the pre-grinding gate.

The alignment response depends on the directional stiffness, contact geometry, lever direction, robot configuration, and compliance of the tool mounting. The experiments therefore vary the selected controller parameters while keeping the calibrated geometry, the robot configuration, and the tool mounting unchanged. A commanded tool orientation offset represents the initial angular misalignment between the tool and the surface. Chapters 2 and 3 introduce the frame conventions and controller formulation used in the experiments.

1.4 Scope and Contributions

The scope covers contact set-up against one calibrated plane. Sustained grinding and adaptive selection of the compliance-centre position are outside the experimental evaluation. The main contributions are the following.

- A real-time Cartesian impedance controller with surface-relative directional gains and a shiftable virtual CoC was implemented on a Franka Emika Panda.
- The influence of directional stiffness and tangential compliance-centre position on

contact-induced end-effector rotation was characterised experimentally.

- A projected null-space controller combining joint damping and singular-value conditioning was implemented and evaluated separately in Cartesian pose hold.

The contact study distinguishes a direction-independent fixed reference from a condition-dependent displaced centre. The former can be selected before the sign and tangent-plane direction of the initial misalignment are known. The latter adds rotational authority through the commanded moment $r_c \times f$ when the required direction is available.

1.5 Thesis Structure

Chapter 2 establishes the impedance law, frame conventions, and compliance-centre shift. Their real-time implementation on the Franka Emika Panda is described in Chapter 3. The experimental method and results follow in Chapters 4 and 5. Chapter 6 states the resulting conclusions and future work.

Chapter 2

Theoretical Background

This chapter presents the mathematical basis of the Cartesian impedance controller. The required frame conventions and pose errors are introduced first. The subsequent sections derive the kinematic mappings, the impedance wrench F , the compliance-centre transformation, and the null-space torque τ_{null} .

2.1 Reference Frames

The following right-handed coordinate frames are used in the theoretical formulation and the controller [21, 3].

- **Base frame $\{0\}$:** The base frame is fixed to the robot mounting surface. Its origin is the robot base reference point used by the kinematic model. For the upright Panda configuration used in this thesis, the z_0 -axis is normal to the mounting surface, while the x_0 - and y_0 -axes lie in the mounting plane. Unless stated otherwise, Cartesian positions, directions, velocities, forces, and moments are expressed in this frame.
- **Joint frames $\{1\}, \{2\}, \dots, \{n\}$:** The joint frames follow the robot kinematic convention and are required for forward kinematics and Jacobian construction; the real-time controller obtains the corresponding quantities from the libfranka model.
- **End-effector frame $\{EE\}$:** The end-effector frame is attached to the controlled TCP and moves with the tool. In the implemented robot model, the end-effector origin and TCP are identified, so $p_{\text{TCP}} = p_{\text{EE}}$. The physical tool face is represented separately by its calibrated geometric offset from this frame. The position p_{EE}

and orientation R_{EE} , both expressed relative to the base frame, define the current Cartesian pose of the robot. The axes x_{EE} , y_{EE} , and z_{EE} describe the current end-effector orientation. The configured tool approach axis is defined in this frame and is transformed to the base frame using R_{EE} .

- **Desired end-effector frame $\{d\}$:** The desired frame represents the commanded TCP pose. Its origin is p_d , and its orientation is R_d , both expressed relative to the base frame. The difference between the desired and current end-effector frames provides the translational and rotational errors used by the impedance controller.
- **Surface frame $\{S\}$:** The surface frame is tied to the configured surface rather than to the moving tool. Its origin is placed at the configured surface point p_s . Its orientation R_{surface} is defined by the two surface tangents t_1 and t_2 and the surface normal n_s :

$$R_{\text{surface}} = \begin{bmatrix} t_1 & t_2 & n_s \end{bmatrix}. \quad (2.1)$$

The directions t_1 and t_2 lie in the surface plane and are mutually orthogonal. The unit normal n_s is defined to point outward from the surface towards the free-space region above the plate. This sign convention is kept throughout the contact sequence, so motion towards the surface is along $-n_s$. The surface frame is used to describe the commanded tool orientation offset and to assign stiffness and damping values according to the two tangential directions and the normal direction.

Figure 2.1 shows these frames together with the pose error between the current and the desired end-effector frame.

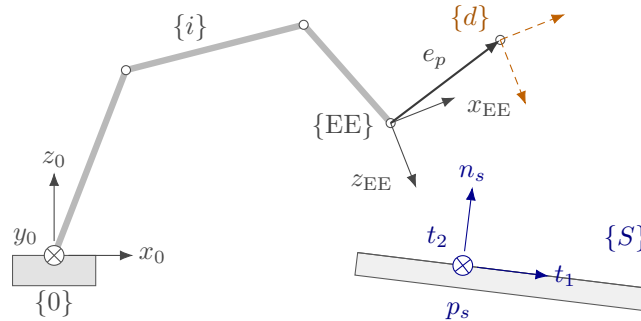


Figure 2.1: Frames used in the controller and the end-effector pose error.

2.2 Rigid-Body Pose and Pose Error

Unless stated otherwise, Cartesian quantities are expressed in the robot base frame. A Cartesian pose consists of a position $p \in \mathbb{R}^3$ and an orientation $R \in SO(3)$, collected in the homogeneous transformation T :

$$T = \begin{bmatrix} R & p \\ 0^\top & 1 \end{bmatrix}. \quad (2.2)$$

The measured end-effector pose is therefore

$$T_{EE} = \begin{bmatrix} R_{EE} & p_{EE} \\ 0^\top & 1 \end{bmatrix}. \quad (2.3)$$

The controller obtains T_{EE} directly from the libfranka robot state. Rotation matrices are used throughout; the pose is not converted to Euler angles.

2.2.1 Cartesian Pose Error

The desired end-effector transformation T_d is

$$T_d = \begin{bmatrix} R_d & p_d \\ 0^\top & 1 \end{bmatrix}, \quad (2.4)$$

where p_d and R_d are the desired TCP position and orientation, respectively. The current end-effector pose is denoted by T_{EE} .

The relative pose-error transformation T_{err} maps the current end-effector frame to the desired frame:

$$T_{err} = T_{EE}^{-1} T_d = \begin{bmatrix} R_{EE}^\top R_d & R_{EE}^\top (p_d - p_{EE}) \\ 0^\top & 1 \end{bmatrix}. \quad (2.5)$$

When the current and desired poses coincide, $T_{err} = I_4$. The rotational and translational blocks of Equation (2.5) therefore both represent zero error in this case.

Position error. The translational block of Equation (2.5) expresses the position error e_p in the current end-effector frame. The position error e_p used by the Cartesian impedance law is expressed in the base frame:

$$e_p = p_d - p_{EE} \in \mathbb{R}^3. \quad (2.6)$$

A positive position error e_p therefore points from the current TCP position towards the desired TCP position.

Orientation error. The relative rotation ΔR is the rotational block of Equation (2.5):

$$\Delta R = R_{EE}^\top R_d. \quad (2.7)$$

It represents the rotation required to carry the current end-effector orientation to the desired orientation. If both orientations coincide, then $\Delta R = I$.

For use in the impedance controller, ΔR is represented by a rotation angle ϕ and a unit rotation axis $u \in \mathbb{R}^3$ [21, 17]. The rotation-matrix and axis-angle representations can be converted in both directions. Given a unit axis u and an angle ϕ , the corresponding rotation matrix is obtained from Rodrigues' formula,

$$R(u, \phi) = I + \sin \phi [u]_\times + (1 - \cos \phi) [u]_\times^2. \quad (2.8)$$

For the relative rotation considered here, $R(u, \phi) = \Delta R$. The skew-symmetric matrix of the rotation axis is

$$[u]_\times = \begin{bmatrix} 0 & -u_z & u_y \\ u_z & 0 & -u_x \\ -u_y & u_x & 0 \end{bmatrix}, \quad [u]_\times^\top = -[u]_\times. \quad (2.9)$$

Conversely, when the rotation matrix ΔR is known, the angle follows from

$$\text{tr}(\Delta R) = 1 + 2 \cos \phi, \quad (2.10)$$

and therefore

$$\phi = \arccos\left(\frac{\text{tr}(\Delta R) - 1}{2}\right), \quad \phi \in [0, \pi]. \quad (2.11)$$

Subtracting the transpose of ΔR isolates its skew-symmetric part:

$$\frac{1}{2}(\Delta R - \Delta R^\top) = \sin \phi [u]_\times. \quad (2.12)$$

For $\sin \phi \neq 0$, the rotation axis is obtained from

$$u = \frac{1}{2 \sin \phi} \begin{bmatrix} \Delta R_{32} - \Delta R_{23} \\ \Delta R_{13} - \Delta R_{31} \\ \Delta R_{21} - \Delta R_{12} \end{bmatrix}. \quad (2.13)$$

Multiplying the unit axis by the angle gives the current-frame orientation error $e_{R,EE}$. Rotating it into the base frame gives the controller orientation error e_R :

$$e_{R,EE} = \phi u, \quad e_R = R_{EE} e_{R,EE} \in \mathbb{R}^3. \quad (2.14)$$

The direction of e_R defines the axis of the required correction, while its magnitude is the angular error in radians. When the current and desired orientations coincide, $\phi = 0$ and $e_R = 0$. Together, $e_p = p_d - p_{EE}$ and e_R are expressed in the base frame and point in the translational and rotational directions required to reduce the pose error.

The complete Cartesian error e stacks the position error e_p and orientation error e_R used by the impedance controller:

$$e = \begin{bmatrix} e_p \\ e_R \end{bmatrix} \in \mathbb{R}^6. \quad (2.15)$$

2.3 Kinematics of a 7-DOF Manipulator

Having defined the end-effector pose as the controlled variable, its relation to the joint configuration is introduced next.

For the joint configuration $q \in \mathbb{R}^7$, forward kinematics defines the end-effector pose $T_{EE}(q)$. The Denavit–Hartenberg convention fixes the joint and frame indexing [4, 3], but the

real-time controller obtains the pose and geometric Jacobian from the libfranka model interface.

The geometric Jacobian $J(q) \in \mathbb{R}^{6 \times 7}$ maps joint velocity to the base-frame end-effector twist,

$$\begin{bmatrix} \dot{p}_{\text{EE}} \\ \omega_{\text{EE}} \end{bmatrix} = J(q)\dot{q}. \quad (2.16)$$

Its transpose maps the commanded Cartesian wrench to joint torque. These two mappings are the kinematic relations required by the impedance and null-space controllers.

2.4 Cartesian Impedance Control

The geometric Jacobian $J(q)$ is also used through its transpose to map a Cartesian wrench F to joint torques τ [13]. Before introducing this mapping, the desired Cartesian interaction behaviour is defined.

In this thesis, Cartesian impedance control is formulated as a spring–damper relation between the end-effector motion and the commanded wrench [11, 19, 15]. The Cartesian pose error e , defined in Section 2.2.1, produces a restoring wrench through the stiffness matrix K , while the Cartesian velocity $\dot{x}$ produces an opposing wrench through the damping matrix D . Their entries determine the translational and rotational compliance in the selected compliance directions [20]. Physically, the stiffness terms determine how strongly a pose error generates a restoring force or moment, while the damping terms oppose the corresponding Cartesian motion.

2.4.1 Impedance Wrench

The impedance wrench is evaluated in the robot base frame. The corresponding translational stiffness and damping matrices are $K_{p,0}, D_{p,0} \in \mathbb{R}^{3 \times 3}$, while the rotational matrices are $K_{R,0}, D_{R,0} \in \mathbb{R}^{3 \times 3}$. These matrices are not generally diagonal in the base frame because the direction-dependent gains are defined relative to the surface frame. Their base-frame representation therefore changes with the surface-frame orientation, even when the local gain values remain unchanged.

The commanded Cartesian force f is given by the translational spring–damper law [11, 15]

$$f = K_{p,0}e_p + D_{p,0}(\dot{p}_d - \dot{p}_{EE}) \in \mathbb{R}^3, \quad (2.17)$$

where $\dot{p}_d$ and $\dot{p}_{EE}$ are the desired and measured linear velocities expressed in the base frame. For a fixed position setpoint, $\dot{p}_d = \mathbf{0}$, and the damping term becomes $-D_{p,0}\dot{p}_{EE}$.

The commanded Cartesian moment m is

$$m = K_{R,0}e_R - D_{R,0}\omega_{EE} \in \mathbb{R}^3, \quad (2.18)$$

where ω_{EE} is the measured end-effector angular velocity expressed in the base frame. Since no angular reference velocity is commanded, the rotational damping acts directly against ω_{EE} .

The Cartesian wrench F stacks the force and moment:

$$F = \begin{bmatrix} f \\ m \end{bmatrix} = \begin{bmatrix} K_{p,0}e_p + D_{p,0}(\dot{p}_d - \dot{p}_{EE}) \\ K_{R,0}e_R - D_{R,0}\omega_{EE} \end{bmatrix} \in \mathbb{R}^6. \quad (2.19)$$

The wrench is ordered as force followed by moment, and all its components are expressed in the base frame. It is mapped to joint torques τ_{cart} through the Jacobian transpose in the following subsection.

Equation (2.19) defines the impedance about the TCP, with translation and rotation decoupled. Section 2.7 extends the same law to an impedance defined about a displaced centre of compliance.

2.4.2 Joint-Torque Control Law

By the principle of virtual work, the commanded Cartesian wrench F is mapped to joint torques τ_{cart} through the transpose of the geometric Jacobian $J(q)$ [13]:

$$\tau_{\text{cart}} = J^\top(q)F, \quad (2.20)$$

where $\tau_{\text{cart}} \in \mathbb{R}^n$ is the Cartesian impedance torque contribution and $J^\top(q) \in \mathbb{R}^{n \times 6}$. Sub-

stituting Equation (2.19) gives

$$\tau_{\text{cart}} = J^\top(q) \begin{bmatrix} K_{p,0}e_p + D_{p,0}(\dot{p}_d - \dot{p}_{\text{EE}}) \\ K_{R,0}e_R - D_{R,0}\omega_{\text{EE}} \end{bmatrix}. \quad (2.21)$$

The gain matrices in this expression are the base-frame matrices obtained from the selected surface-coordinate gains. A change in the surface-frame orientation therefore changes the Cartesian wrench F and the resulting joint-torque contribution τ_{cart} , even when the local gain values remain unchanged.

2.4.3 Coriolis and Gravity Compensation in libfranka

In a general model-compensated torque controller, the commanded joint torque τ_{cmd} combines Cartesian impedance, gravity, and Coriolis/centrifugal compensation:

$$\tau_{\text{cmd}} = \tau_{\text{cart}} + \tau_g(q) + \tau_c(q, \dot{q}), \quad (2.22)$$

where $\tau_g(q)$ is the gravity torque and $\tau_c(q, \dot{q})$ is the Coriolis and centrifugal torque.

These terms are handled differently by the Franka Control Interface. The Coriolis vector provided by the libfranka model is added explicitly to the command because it is not compensated internally. A separate gravity term is not added. In external joint-torque mode, the robot compensates gravity and motor friction internally, and the commanded torque is interpreted as excluding gravity [7]. Adding $\tau_g(q)$ would therefore compensate gravity twice.

The implemented commanded torque $\tau_{\text{cmd,impl}}$ is consequently

$$\tau_{\text{cmd,impl}} = \tau_{\text{cart}} + \tau_c(q, \dot{q}) = J^\top(q)F + \tau_c(q, \dot{q}), \quad (2.23)$$

which follows the official libfranka Cartesian impedance example [8]. The null-space torque τ_{null} introduced later is added to this expression to obtain the complete commanded torque in Equation (2.66).

2.5 Surface Frame

The surface frame provides the tangential and normal directions used for the Cartesian controller quantities. Its orientation is constructed from the unit surface normal n_s , expressed in the robot base frame. The sign of n_s is chosen so that it points outward from the surface towards the free-space region above the plate, which makes $-n_s$ the approach and set-up direction.

The first surface tangent is chosen to follow the base-frame x_0 -direction as closely as possible while remaining in the surface plane. Since the surface is not exactly parallel to the base x_0y_0 -plane, e_{x_0} is projected onto the surface plane and normalised:

$$t_1 = \frac{e_{x_0} - n_s (n_s^\top e_{x_0})}{\|e_{x_0} - n_s (n_s^\top e_{x_0})\|}. \quad (2.24)$$

The second tangent completes the right-handed frame,

$$t_2 = n_s \times t_1. \quad (2.25)$$

The construction requires e_{x_0} not to be parallel to n_s , and t_1 and t_2 are then orthogonal unit vectors in the surface plane. Because the plane and its normal do not depend on that reference direction, a different in-plane reference would rotate t_1 and t_2 within the surface without changing the surface geometry, so an axis-specific result holds with respect to the direction that was configured, the first-tangent hint recorded in Appendix C. The tangents obtained for the reported experiments are given in Section 4.2.

The orientation of the surface frame relative to the robot base frame is

$$R_{\text{surface}} = \begin{bmatrix} t_1 & t_2 & n_s \end{bmatrix}. \quad (2.26)$$

Surface-frame quantities therefore use the coordinate order

$$\begin{bmatrix} t_1 & t_2 & n_s \end{bmatrix}^\top.$$

This convention is used consistently for the directional Cartesian quantities. For example, K_{p,t_1} , K_{p,t_2} , and $K_{p,n}$ denote the translational stiffness entries along the two surface tan-

gents and the surface normal, respectively. The same ordering is used for the rotational gains and for surface-frame components of the compliance-centre displacement.

2.6 Stiffness and Damping Representation

The Cartesian impedance law of Section 2.4 is evaluated in the robot base frame. The directional stiffness and damping values are specified relative to the calibrated surface, because the relevant directions for the contact problem are the two surface tangents t_1 and t_2 and the surface normal n_s .

The translational and rotational stiffness and damping matrices are first defined in surface coordinates as

$$K_{p,S} = \begin{bmatrix} K_{p,t_1} & 0 & 0 \\ 0 & K_{p,t_2} & 0 \\ 0 & 0 & K_{p,n} \end{bmatrix}, \quad D_{p,S} = \begin{bmatrix} D_{p,t_1} & 0 & 0 \\ 0 & D_{p,t_2} & 0 \\ 0 & 0 & D_{p,n} \end{bmatrix}, \quad (2.27)$$

$$K_{R,S} = \begin{bmatrix} K_{R,t_1} & 0 & 0 \\ 0 & K_{R,t_2} & 0 \\ 0 & 0 & K_{R,n} \end{bmatrix}, \quad D_{R,S} = \begin{bmatrix} D_{R,t_1} & 0 & 0 \\ 0 & D_{R,t_2} & 0 \\ 0 & 0 & D_{R,n} \end{bmatrix}. \quad (2.28)$$

The common axis order is $[t_1, t_2, n_s]$.

For use in the base-frame impedance law, these matrices are rotated into the robot base frame:

$$K_{p,0} = R_{\text{surface}} K_{p,S} R_{\text{surface}}^\top, \quad D_{p,0} = R_{\text{surface}} D_{p,S} R_{\text{surface}}^\top, \quad (2.29)$$

$$K_{R,0} = R_{\text{surface}} K_{R,S} R_{\text{surface}}^\top, \quad D_{R,0} = R_{\text{surface}} D_{R,S} R_{\text{surface}}^\top. \quad (2.30)$$

The complete base-frame stiffness and damping matrices are then

$$K_0 = \begin{bmatrix} K_{p,0} & 0 \\ 0 & K_{R,0} \end{bmatrix}, \quad D_0 = \begin{bmatrix} D_{p,0} & 0 \\ 0 & D_{R,0} \end{bmatrix}. \quad (2.31)$$

At this stage translation and rotation remain decoupled. The centre-of-compliance transformation introduced in Section 2.7 can subsequently introduce coupling between the two blocks.

The transformation changes only the coordinate representation of the gain matrices. The physical directions associated with the gains remain tied to the calibrated surface. For example, $K_{p,n}$ acts along n_s , even when the surface normal is not aligned with a robot-base axis.

For the reported contact experiments, the directional gain values were assigned along $[t_1, t_2, n_s]$ and then transformed to the base frame before evaluating the Cartesian impedance law. The gains therefore remained fixed relative to the calibrated surface and did not rotate with the current end-effector orientation R_{EE} . The phase-indexed numerical entries are given in Table 4.2.

2.7 Centre of Compliance

The surface-frame transformations of Equations (2.29) and (2.30) change the directions along which the Cartesian stiffness and damping act, while leaving their reference point at the tool centre point (TCP). Independently of that orientation choice, the Cartesian impedance can be defined about a virtual CoC p_c . Shifting this point away from the TCP introduces translation–rotation coupling into the Cartesian impedance [17, 5, 22, 14]. The construction is applied during contact set-up, where it lets the normal pressing action generate a rotational moment as well as a translational contact load.

For a dual-arm assembly task, Suomalainen et al. experimentally showed that compliance-centre placement and compliant-axis selection affect the alignment response [23]. This experimental result motivates the placement study in this thesis. The point-shift adjoint below remains derived from the spatial-transformation formulation.

The displacement from the TCP to the CoC is

$$r_c = p_c - p_{\text{TCP}} \in \mathbb{R}^3, \quad p_{\text{TCP}} = p_{EE}. \quad (2.32)$$

The vector r_c describes only the virtual shift of the impedance reference point.

Let $[r_c]_\times$ denote the skew-symmetric matrix satisfying $[r_c]_\times a = r_c \times a$. Under the local small-angle formulation, the translational error referred to the CoC is

$$e_{p,c} = e_p - [r_c]_\times e_R. \quad (2.33)$$

The negative sign follows from the selected lever convention. The vector from the TCP to p_c is r_c itself, and therefore

$$e_R \times r_c = -[r_c]_\times e_R.$$

The stacked error at the CoC collects $e_{p,c}$ and the rotational error e_R . The point-shift adjoint $\text{Ad}(r_c)$ maps the corresponding TCP error onto it:

$$e_c = \begin{bmatrix} e_{p,c} \\ e_R \end{bmatrix} = \text{Ad}(r_c) \begin{bmatrix} e_p \\ e_R \end{bmatrix}, \quad \text{Ad}(r_c) = \begin{bmatrix} I_3 & -[r_c]_\times \\ 0 & I_3 \end{bmatrix}. \quad (2.34)$$

The surface-frame rotation and the point shift have different functions. The transformations in Equations (2.29) and (2.30) change the principal directions of the impedance. The point-shift adjoint $\text{Ad}(r_c)$ changes the point about which the impedance is defined. After the gains have been expressed in the base frame, the translational and rotational stiffness and damping matrices are assembled at the CoC as

$$K_c = \begin{bmatrix} K_p & 0 \\ 0 & K_R \end{bmatrix}, \quad D_c = \begin{bmatrix} D_p & 0 \\ 0 & D_R \end{bmatrix}. \quad (2.35)$$

All blocks in Equation (2.35) are expressed in the base-frame orientation. The compliance-centre wrench is $F_c = K_c e_c$, and transforming this power-conjugate wrench gives the corresponding TCP wrench $F_{\text{TCP}} = \text{Ad}(r_c)^\top F_c$. The matrices at the TCP therefore follow from the congruence transformations

$$K_{\text{TCP}} = \text{Ad}(r_c)^\top K_c \text{Ad}(r_c) = \begin{bmatrix} K_p & -K_p[r_c]_\times \\ -[r_c]_\times^\top K_p & K_R + [r_c]_\times^\top K_p[r_c]_\times \end{bmatrix} \quad (2.36)$$

and

$$D_{\text{TCP}} = \text{Ad}(r_c)^\top D_c \text{Ad}(r_c) = \begin{bmatrix} D_p & -D_p[r_c]_\times \\ -[r_c]_\times^\top D_p & D_R + [r_c]_\times^\top D_p [r_c]_\times \end{bmatrix}. \quad (2.37)$$

The off-diagonal blocks introduce translation–rotation coupling. A translational position or velocity error can therefore contribute to the commanded moment, while a rotational error or velocity can contribute to the commanded force. For $r_c = 0$ the point shift disappears and the block-diagonal impedance at the TCP is recovered.

2.7.1 Commanded Moment

The Cartesian wrench generated by the impedance controller is $F = [f^\top, m^\top]^\top$, where f and m denote the complete commanded force and moment at the TCP.

Without a CoC displacement, the commanded moment is given by the ordinary rotational impedance,

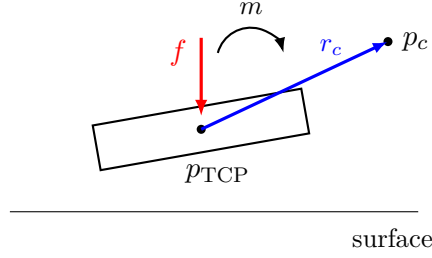
$$m_R = K_R e_R - D_R \omega_{EE}. \quad (2.38)$$

A non-zero CoC displacement adds the translation–rotation coupling contribution $r_c \times f$, generated by the complete commanded force. The complete commanded TCP moment is therefore

$$m = m_R + r_c \times f. \quad (2.39)$$

This relation is exact for the point-shifted stiffness and damping matrices above. The controller evaluates the complete shifted spring–damper wrench and does not separately command an elastic or a damping coupling moment.

For $r_c = 0$ the coupling contribution vanishes and $m = m_R$. The TCP-centred condition is therefore a zero-coupling condition rather than a zero-moment condition. The rotational compliance remains finite and the commanded rotational impedance remains active.



$$m = m_R + r_c \times f$$

Figure 2.2: Centre-of-compliance displacement and commanded TCP moment. Shifting the virtual centre from the TCP by r_c introduces the coupling contribution $r_c \times f$.

2.7.2 Surface-Frame and Tangential Compliance-Centre Displacement

For surface contact the CoC displacement is resolved along the two surface tangents and the surface normal. Its coordinates in the surface frame are

$$r_{c,S} = \begin{bmatrix} r_{c,t_1} \\ r_{c,t_2} \\ r_{c,n} \end{bmatrix}. \quad (2.40)$$

Since $R_{\text{surface}} = [t_1 \ t_2 \ n_s]$ holds those surface axes expressed in the base frame, the same physical displacement in base coordinates is

$$r_{c,0} = R_{\text{surface}} r_{c,S} = r_{c,t_1} t_1 + r_{c,t_2} t_2 + r_{c,n} n_s, \quad (2.41)$$

with the inverse relation $r_{c,S} = R_{\text{surface}}^\top r_{c,0}$. The tangential part collects the two in-plane components,

$$r_{c,t} = r_{c,t_1} t_1 + r_{c,t_2} t_2, \quad (2.42)$$

while $r_{c,n} n_s$ is the part along the surface normal. The quantities have to be kept apart: r_{c,t_1} and r_{c,t_2} are signed scalar coordinates, $r_{c,t}$ is the tangential vector they form, and its magnitude is written $\|r_{c,t}\|$. For a purely tangential displacement $r_{c,n} = 0$, and therefore $\|r_c\| = \|r_{c,t}\|$.

A non-zero displacement is specified by the frame in which its components are held con-

stant as well as by its magnitude and direction. The two definitions follow the same pattern and differ only in the rotation that carries the configured vector into the base frame:

$$r_{c,0} = R_{\text{surface}} r_{c,S}, \quad r_{c,0} = R_{\text{EE}} r_{c,\text{EE}}. \quad (2.43)$$

For a surface-fixed definition $r_{c,S}$ remains constant, and because the surface frame is fixed for one controller session the base-frame displacement is constant during set-up. For a tool-fixed definition $r_{c,\text{EE}}$ remains constant instead, so $r_{c,0}$ rotates with the end effector. Section 3.2.4 describes both implementations.

The commanded force enters through $r_c \times f$, and it decomposes exactly into a normal and a tangential component,

$$f = f_n + f_t, \quad f_n = F_n n_s, \quad f_t = f - f_n, \quad (2.44)$$

with $F_n = n_s^\top f$ the signed commanded normal force. The normal component is the projection of f onto n_s , and the tangential component is what remains of f in the surface plane. The complete CoC contribution is therefore

$$r_c \times f = r_c \times f_n + r_c \times f_t. \quad (2.45)$$

Both terms remain part of the complete commanded moment m .

The normal-press term simplifies. Substituting $r_c = r_{c,t} + r_{c,n} n_s$ and using $f_n \parallel n_s$, the normal component of the displacement drops out, $r_{c,n} n_s \times f_n = \mathbf{0}$, which leaves

$$r_c \times f_n = r_{c,t} \times f_n. \quad (2.46)$$

Only the tangential part of the displacement produces a moment from the commanded normal press. A displacement along the surface normal contributes nothing to this term. Consequently, the moment generated by the normal press depends on the tangential projection $r_{c,t}$, not on the total distance $\|r_c\|$ between the TCP and the CoC.

The right-handed surface frame satisfies

$$t_1 \times t_2 = n_s, \quad t_2 \times n_s = t_1, \quad t_1 \times n_s = -t_2. \quad (2.47)$$

Let the commanded orientation offset in the surface plane be

$$\theta_{\text{cmd}} = \theta_{t_1} t_1 + \theta_{t_2} t_2. \quad (2.48)$$

For a prescribed tangential magnitude $\|r_{c,t}\|$, the displacement used to assist the commanded alignment rotation is chosen perpendicular to the orientation-offset direction,

$$r_{c,t} = \|r_{c,t}\| \frac{\theta_{t_1} t_2 - \theta_{t_2} t_1}{\sqrt{\theta_{t_1}^2 + \theta_{t_2}^2}}. \quad (2.49)$$

The choice fixes the direction; the magnitude remains a free parameter of the configuration. Using Equation (2.47), the normal-press contribution becomes

$$r_{c,t} \times f_n = \|r_{c,t}\| F_n \frac{\theta_{t_1} t_1 + \theta_{t_2} t_2}{\sqrt{\theta_{t_1}^2 + \theta_{t_2}^2}}. \quad (2.50)$$

During the set-up press $F_n < 0$, so this contribution acts opposite to θ_{cmd} . For the two principal directions the rule reduces to

$$r_{c,t} = +\|r_{c,t}\| t_2 \quad \text{for } +\theta_{t_1}, \quad r_{c,t} = -\|r_{c,t}\| t_1 \quad \text{for } +\theta_{t_2}, \quad (2.51)$$

giving $r_{c,t} \times f_n = \|r_{c,t}\| F_n t_1$ and $r_{c,t} \times f_n = \|r_{c,t}\| F_n t_2$ respectively. Reversing the orientation offset reverses $r_{c,t}$ and therefore reverses this contribution. Figure 2.3 shows the geometry for the two signs.

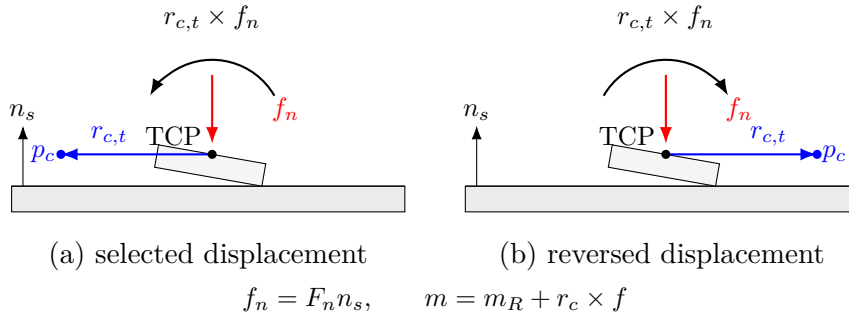


Figure 2.3: Normal-force contribution to the compliance-centre coupling moment for opposite tangential displacements: (a) the displacement selected from the orientation offset; (b) the displacement with the opposite sign.

The complete commanded moment remains $m = m_R + r_c \times f$. The normal-force decomposition is used only to explain the directional effect of the CoC displacement. Tangential commanded-force components and damping effects remain contained in the complete wrench generated by the controller.

The coupling vanishes for $r_c = 0$. Conversely, if K_p is nonsingular, zero coupling implies $r_c = 0$. Since $\text{Ad}(r_c)$ is invertible, the congruence transformation also preserves definiteness:

$$K_c \succeq 0 \implies K_{\text{TCP}} \succeq 0, \quad K_c \succ 0 \implies K_{\text{TCP}} \succ 0. \quad (2.52)$$

The same statements apply to D_c and D_{TCP} . Negative off-diagonal entries therefore represent coupling terms rather than negative stiffness directions. Closed-loop robot–controller–contact behaviour with phase switching was assessed experimentally under the operating conditions reported in Chapter 4.

This formulation is a local small-motion model in which the displacement is treated as fixed during one controller evaluation. A surface-fixed $r_{c,S}$ therefore keeps $r_{c,0}$ constant during set-up, whereas a tool-fixed displacement does not, and the controller updates the base-frame vector and the shifted gain matrices during each control cycle in that case.

The virtual CoC p_c is consequently a selected impedance reference point and is not interpreted as a measured physical pivot. It need not coincide with the physical contact location. The resulting contact response additionally depends on the finite rotational stiffness, the contact constraint, friction, the robot configuration, and tool-mount compliance.

2.8 Null-Space Torque

The Panda has one redundant joint direction when its 6×7 geometric Jacobian $J(q)$ has full row rank. The singular value decomposition (SVD) supplies two different quantities: v_7 spans the redundant direction, while $\sigma_{\min} = \sigma_6$ indicates the weakest task-producing mapping. A thresholded pseudoinverse gives the velocity and torque projectors. The controller then combines projected joint damping with a two-probe conditioning torque.

2.8.1 SVD and Null-Space Direction

The decomposition of the current Jacobian is

$$J(q) = U_J \Sigma_J V_J^\top, \quad (2.53)$$

where the six task-producing singular values satisfy $\sigma_1 \geq \dots \geq \sigma_6 \geq 0$. At full row rank, the seventh right-singular vector satisfies

$$J v_7 = 0, \quad (2.54)$$

and spans the one-dimensional kinematic null space. The conditioning indicator is

$$\sigma_{\min} = \sigma_6. \quad (2.55)$$

The vector v_7 and scalar σ_6 therefore have different roles. The first defines redundant motion; the second describes the weakest of the six task mappings. Because the geometric Jacobian mixes linear and angular rows, $\sigma_{\min}$ is used only as a relative indicator under one fixed Jacobian convention.

2.8.2 Pseudoinverse and Projectors

A relative tolerance ρ defines the retained singular values by $\sigma_i > \rho \sigma_1$. The corresponding Moore–Penrose pseudoinverse is

$$J^+ = \sum_{\sigma_i > \rho \sigma_1}^6 \frac{1}{\sigma_i} v_i u_i^\top. \quad (2.56)$$

The joint-velocity projector extracts redundant motion,

$$N_q = I_7 - J^+ J, \quad \dot{q}_{\text{null}} = N_q \dot{q}, \quad (2.57)$$

while the transpose projector applies a secondary joint torque without a first-order task component,

$$N_\tau = N_q^\top = I_7 - J^\top J^{+\top}. \quad (2.58)$$

With the thresholded SVD construction, the projector is orthogonal and symmetric. At full row rank it reduces to $N_q = N_\tau = v_7 v_7^\top$. The implemented separation is kinematic rather than inertia weighted; Cartesian position retention is therefore checked in the pose-hold experiment.

2.8.3 Projected Damping and Conditioning

Projected null-space damping opposes the redundant joint velocity,

$$\tau_d = -d_{\text{null}} N_\tau \dot{q}. \quad (2.59)$$

The conditioning term samples two configurations along the current redundant direction. For the probe angle α_{probe} , these are

$$q_+ = q + \alpha_{\text{probe}} v_7, \quad q_- = q - \alpha_{\text{probe}} v_7. \quad (2.60)$$

Their minimum singular values are

$$\sigma_+ = \sigma_{\min}(J(q_+)), \quad \sigma_- = \sigma_{\min}(J(q_-)), \quad (2.61)$$

with the sampled difference

$$\Delta_\sigma = \sigma_+ - \sigma_-. \quad (2.62)$$

The direction selector applies a deadband δ_σ ,

$$s_\sigma = \begin{cases} +1, & \Delta_\sigma > \delta_\sigma, \\ -1, & \Delta_\sigma < -\delta_\sigma, \\ 0, & |\Delta_\sigma| \leq \delta_\sigma, \end{cases} \quad (2.63)$$

and the conditioning torque is

$$\tau_\sigma = k_\sigma N_\tau(s_\sigma v_7). \quad (2.64)$$

Sampling q_+ and q_- selects the neighbouring configuration with the larger $\sigma_{\min}$. The finite probes are local tangent approximations to the nonlinear self-motion. The commanded magnitude k_σ is independent of the probe angle.

2.8.4 Complete Null-Space Torque

The complete secondary torque is

$$\tau_{\text{null}} = \tau_d + \tau_\sigma = -d_{\text{null}} N_\tau \dot{q} + k_\sigma N_\tau(s_\sigma v_7). \quad (2.65)$$

The damping term acts only while redundant motion is present. The conditioning term is an active configuration objective and can act at zero joint velocity. Its direction is invariant to the arbitrary sign returned for v_7 , because reversing v_7 also exchanges the two probes and reverses s_σ .

2.9 Complete Joint-Torque Command

The complete joint-torque command τ_{cmd} combines the Cartesian impedance torque τ_{cart} , the null-space torque τ_{null} , and the Coriolis compensation τ_c :

$$\tau_{\text{cmd}} = \tau_{\text{cart}} + \tau_{\text{null}} + \tau_c(q, \dot{q}) = J^\top(q)F + \tau_{\text{null}} + \tau_c(q, \dot{q}). \quad (2.66)$$

Here, $\tau_{\text{cart}} = J^\top(q)F$ is the Cartesian impedance torque, τ_{null} is the secondary null-space torque, and $\tau_c(q, \dot{q})$ is the Coriolis and centrifugal compensation obtained from the robot model and added in the control callback. A separate gravity term is not added because gravity compensation is handled internally by the Franka Control Interface, as described in Section 2.4.3. The next chapter implements these torque contributions in the real-time control loop and combines them with the surface-relative geometry, phase-dependent references, and contact sequence.

Chapter 3

Controller Design and Real-Time Implementation

This chapter describes the real-time implementation of the Cartesian impedance controller developed in Chapter 2. The implementation combines surface-relative geometry, phase-dependent reference generation, a selectable virtual CoC, null-space control, and real-time data handling. Mathematical relations already established in Chapter 2 are referenced here rather than restated, so that this chapter carries only what the controller adds to them. The controller is written in C++ using libfranka and Eigen and is executed through the Franka Control Interface (FCI) at a nominal control period of 1 ms. Numerical settings used in the experiments are given in Chapter 4 and Appendix C.

3.1 Controller Architecture and Real-Time Control Cycle

The software separates configuration and operator interaction from the real-time control calculation and from post-run data processing. The complete parameter set is read and validated before a controller session starts, so the real-time callback receives only the prepared configuration, preallocated state, the robot-model interface, and non-blocking operator requests. File output and run evaluation are performed after torque control has stopped.

At program start the FCI connection is established, error recovery is requested, the selected collision behaviour is applied, and the robot model is loaded. A separate keyboard thread receives operator requests and transfers them through atomic variables, which keeps terminal input outside the real-time callback. Reading and validating the configuration again before each session permits settings to be changed between runs without recompiling.

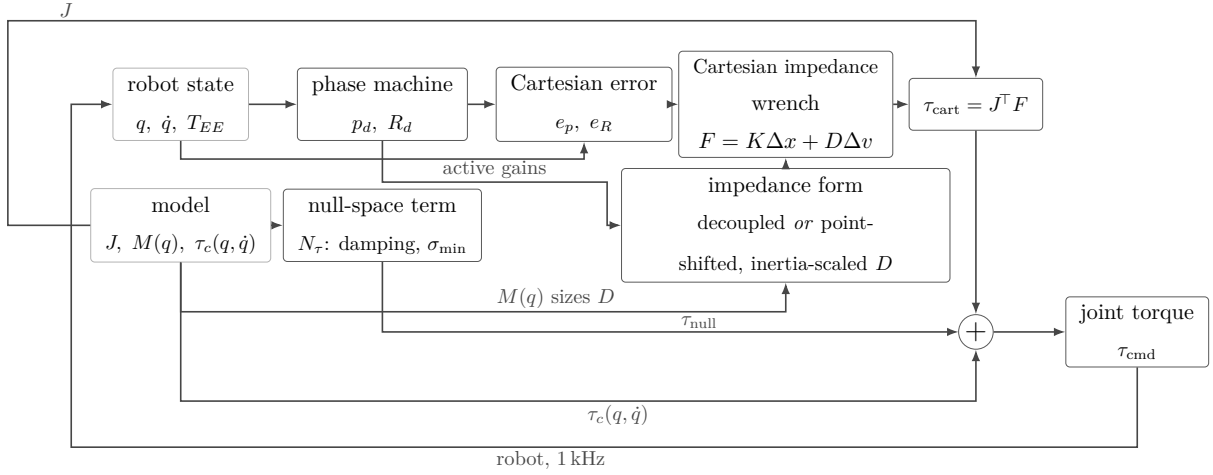


Figure 3.1: Torque contributions combined in one control cycle.

Figure 3.1 shows the torque path of one impedance-controlled cycle, and the order within that cycle is fixed. The callback first reads the measured joint configuration q , the joint velocity $\dot{q}$, the end-effector pose, the 6×7 geometric Jacobian $J(q)$, and the remaining robot-model quantities. The phase logic then updates the desired Cartesian reference. The Cartesian pose error and the impedance wrench $F = [f^\top, m^\top]^\top$ are evaluated according to Section 2.2.1 and Section 2.4, using the reference of the active phase and the impedance matrices selected for it. That wrench is mapped to joint torque through $J^\top(q)$, the secondary null-space torque and the model Coriolis contribution are added, the required signals are stored in preallocated memory, and the resulting torque vector is returned to libfranka.

The end-effector pose supplied by the robot state is used directly for p_{EE} and R_{EE} . The real-time controller does not reconstruct that pose from a separate Denavit–Hartenberg chain.

Two properties of the error evaluation belong to the implementation rather than to the impedance law. A rotational mask can be applied by resolving e_R in the surface frame, retaining the enabled components, and transforming the result back to the base frame;

the mask acts on the rotational spring error alone, while the damping term continues to use the measured angular velocity. The controller also carries two impedance branches, one decoupled and one shifted to a virtual CoC, and the active phase selects between them.

The complete impedance-controlled command is

$$\tau_{\text{cmd}} = J^\top(q)F + \tau_{\text{null}} + \tau_c(q, \dot{q}). \quad (3.1)$$

Here τ_{null} is the secondary null-space torque of Section 2.8.4 and $\tau_c(q, \dot{q})$ is the Coriolis and centrifugal contribution obtained from the libfranka model. No separate gravity term is added, because the FCI compensates gravity for the external joint-torque command, as described in Section 2.4.3. An experiment-specific disturbance torque is added only for the null-space pose-hold experiment of Section 4.6 and is therefore not part of the nominal controller formulation.

The implemented controller is used in two configurations in this thesis. The surface-contact configuration generates a phase-dependent Cartesian reference for tool orientation, surface approach, contact set-up and the subsequent grinding phase. The Cartesian pose-hold configuration instead captures and maintains a fixed Cartesian pose, and it carries the separate null-space investigation of Section 4.6. Both run the callback above, and they differ in the reference they generate and in the impedance they activate. Each is described in full before the other begins.

3.2 Surface-Contact Sequence

A surface-contact run passes through four phases, drawn in Figure 3.2. The step between approach and set-up is a transition rather than a fifth phase. It occurs when the geometric clearance of the tool face reaches its configured threshold. The quantities that initialise the set-up reference are captured at that instant.

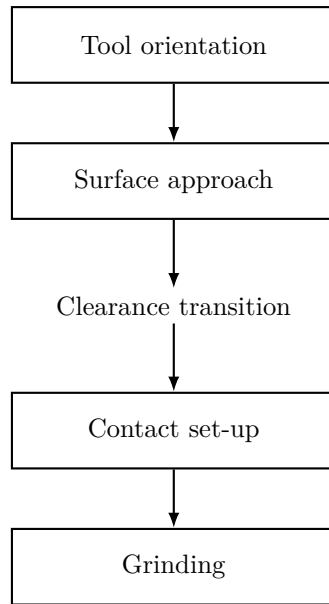


Figure 3.2: Surface-contact phase sequence.

- Tool orientation moves the tool to the commanded surface-relative angular offset while the captured TCP position is held.
- Surface approach retains that orientation and advances the position reference towards the surface along $-n_s$. The phase ends at the clearance transition, where the pose and the selected tool-point offset used to initialise set-up are captured.
- Contact set-up presses the selected tool point onto the surface while the captured orientation is held. Contact-induced end-effector rotation and the optional compliance-centre coupling act in this phase.
- Grinding retains the normal press and superimposes a tangential motion. No reported run entered it.

A phase is a generated Cartesian reference together with an active impedance configuration, and the four fall into two gain groups. Tool orientation and surface approach share the first. Contact set-up uses the compliant contact gains and can activate the shifted virtual CoC, while grinding retains those source gains and returns to the decoupled Cartesian impedance. Section 3.2.4 gives the construction, and the numerical values are stated in Chapter 4. The subsections below follow the sequence in the order the controller executes it.

3.2.1 Surface and Tool Reference

The sequence requires a frame for resolving the surface-relative directions, an orientation target for the tool face, and a geometric representation of the rectangular tool face. These quantities are defined below.

The controller constructs the surface frame according to Section 2.5 from a configured surface point p_s , the surface orientation, and a configured first-tangent hint. That hint is the in-plane reference direction of Section 2.5 and is a parameter rather than a fixed axis, so a fallback direction is applied when the configured one lies close to n_s . The resulting directions t_1 , t_2 and n_s are then used consistently for the descent direction $-n_s$, the surface-relative orientation command, the directional Cartesian gains, the compliance-centre definition when a surface-fixed displacement is selected, and the surface-resolved quantities of the experimental evaluation.

The tool normal is calibrated once in the end-effector frame. Its current direction in the base frame follows from the measured end-effector orientation:

$$n_{\text{Tool},0} = R_{\text{EE}} n_{\text{Tool,EE}}. \quad (3.2)$$

The commanded orientation offset about the two surface tangents is represented by the rotation vector

$$\theta_{\text{cmd}} = \theta_{t_1} t_1 + \theta_{t_2} t_2 \in \mathbb{R}^3. \quad (3.3)$$

Its magnitude $\|\theta_{\text{cmd}}\|$ is the commanded offset angle. For a non-zero offset the corresponding unit rotation axis is

$$u_{\text{cmd}} = \frac{\theta_{\text{cmd}}}{\|\theta_{\text{cmd}}\|}, \quad (3.4)$$

and the axis-angle relation of Equation (2.8) gives the commanded rotation matrix $R_{\text{cmd}} = R(u_{\text{cmd}}, \|\theta_{\text{cmd}}\|)$, which is the identity for a zero commanded offset. This single vector covers the conditions the experiments use: an offset about t_1 alone, an offset about t_2 alone, and the intermediate tangent-plane directions.

Since n_s points out of the surface, the tool-normal direction for a tool face parallel to the

surface is $-n_s$. Applying the commanded orientation offset to that direction gives the desired tool-normal direction

$$n_d = R_{\text{cmd}}(-n_s). \quad (3.5)$$

For a zero commanded offset $n_d = -n_s$. The shortest rotation that aligns the captured tool direction with n_d is applied to the complete start orientation. The implementation can additionally prescribe a twist about the desired tool axis, which fixes the in-plane orientation of the rectangular face; because a centred rectangle is unchanged by a half-turn, the equivalent orientation closest to the captured start pose is selected. The settings used for the measurements are stated in Chapter 4.

Surface approach cannot be decided from the TCP height alone, because a face that is not parallel to the surface reaches the plane first at a corner or along an edge. The tool face is therefore represented as a rectangle fixed in the end-effector frame. Its centre offset, half-width vector, and half-length vector define the four end-effector-frame corner offsets $r_{i,\text{EE}}$,

$$r_{i,\text{EE}} \in \{r_{\text{face,EE}} \pm r_{\text{width,EE}} \pm r_{\text{length,EE}}\}. \quad (3.6)$$

The offsets are defined from the tool centre point, which coincides with the reported end-effector origin in this implementation, $p_{\text{TCP}} = p_{\text{EE}}$. During surface approach the four corners determine the minimum tool clearance and the contact reference used for set-up, as described in Section 3.2.3.

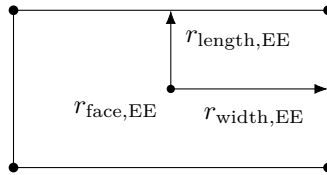


Figure 3.3: Calibrated tool face and its four corners.

3.2.2 Tool Orientation

The tool-orientation phase holds the captured start position while the desired orientation is advanced towards the surface-relative target of Equation (3.5), following a rate-limited

interpolation from the captured start orientation. Its transition is evaluated from the tool-axis error and, when the in-plane twist is enabled, the remaining spin error about the target tool axis. A configured minimum duration prevents immediate acceptance of a transient orientation, and a timeout supplies an alternative transition path. The measured orientation at this transition is the orientation from which descent begins, and it remains the desired orientation throughout the approach that follows.

3.2.3 Surface Approach, Clearance and Tool-Point Selection

Surface approach advances the TCP towards the plane along $-n_s$. Its scalar coordinate $s_{\text{app}}(t)$ is the commanded travel from the TCP position $p_{\text{TCP,start}}$ captured when the phase begins at $t_{\text{app},0}$,

$$s_{\text{app}}(t) = \begin{cases} v_{\text{app}}(t - t_{\text{app},0}), & t_{\text{app},0} \leq t < t_{\text{app,max}}, \\ s_{\text{app,max}}, & t \geq t_{\text{app,max}}, \end{cases} \quad (3.7)$$

with the configured descent speed v_{app} , the configured maximum descent distance $s_{\text{app,max}}$, and $t_{\text{app,max}}$ the instant at which the ramp reaches that distance. Both configured values are listed in Table 4.3, and the saturation bounds an approach that does not reach the clearance threshold. The desired TCP position follows as

$$p_d(t) = p_{\text{TCP,start}} - s_{\text{app}}(t)n_s, \quad (3.8)$$

During the linear approach, the desired TCP velocity is $\dot{p}_d(t) = -v_{\text{app}}n_s$. Reaching $s_{\text{app,max}}$ before the clearance criterion is satisfied terminates the approach as unsuccessful. Where the tool face is not parallel to the surface, the corner closest to the plane can lie well below the TCP, so the TCP height alone does not give the clearance the transition needs. The controller therefore evaluates the four corners of Figure 3.3 in every control cycle. Their positions and clearances also identify the point of the face that set-up should press onto the surface.

The base-frame position $p_i(t)$ of tool-face corner i is

$$p_i(t) = p_{\text{TCP}}(t) + R_{\text{EE}}(t)r_{i,\text{EE}}. \quad (3.9)$$

Its signed clearance $h_i(t)$ from the calibrated surface is

$$h_i(t) = n_s^\top (p_i(t) - p_s). \quad (3.10)$$

Smaller values of h_i place a corner closer to the surface. Since p_{TCP} and p_s are common to all four corners, comparing the clearances h_i is equivalent to comparing the rotated corner offsets along the surface approach direction. The implementation uses this equivalent form for the corner comparison.

The minimum tool clearance $h_{\text{Tool}}(t)$ is the minimum signed clearance between the rectangular tool face and the calibrated surface, evaluated from its four corners:

$$h_{\text{Tool}}(t) = \min_{i \in \{1,2,3,4\}} h_i(t). \quad (3.11)$$

The phase ends at the clearance transition, where

$$h_{\text{Tool}}(t) \leq h_{\text{clearance}}, \quad (3.12)$$

with $h_{\text{clearance}}$ the configured tool clearance of Table 4.3. The approach trajectory is generated for the TCP, whereas the transition to set-up is determined from the physical tool geometry. Since h_{Tool} is the minimum clearance of the four tool-face corners, $h_{\text{clearance}}$ refers to the part of the rectangular tool face closest to the plane rather than to the TCP. For the reported value $h_{\text{clearance}} = 0.020$ m, this part of the tool face is 20 mm above the calibrated plane at the transition. The criterion therefore provides the same specified tool clearance for different tool orientations and is purely geometric.

At the clearance transition, the same four clearances determine the selected tool point used for set-up. A corner belongs to the selected index set $\mathcal{I}_{\text{sel}}$ when its clearance differs from the closest corner by no more than the tool-point selection tolerance ε_{sel} :

$$\mathcal{I}_{\text{sel}} = \{i \in \{1, 2, 3, 4\} \mid h_i - h_{\text{Tool}} \leq \varepsilon_{\text{sel}}\}. \quad (3.13)$$

The tolerance is recorded in Table 4.1. The principal geometric cases contain one selected corner, two neighbouring selected corners, or all four corners. These give the leading corner, the leading-edge midpoint, and the tool-face centre, respectively. With $N = |\mathcal{I}_{\text{sel}}|$, the selected tool-point offset $r_{\text{Tool,EE}}$ is the mean of the selected corner offsets,

$$r_{\text{Tool,EE}} = \frac{1}{N} \sum_{i \in \mathcal{I}_{\text{sel}}} r_{i,\text{EE}}. \quad (3.14)$$

The selected tool point p_{Tool} is the geometric contact reference used during set-up. Its position is

$$p_{\text{Tool}} = p_{\text{TCP}} + R_{\text{EE}} r_{\text{Tool,EE}}. \quad (3.15)$$

The principal geometric cases are shown in Figure 3.4. For the tool-face centre, $r_{\text{Tool,EE}} = r_{\text{face,EE}}$. That calibrated offset lies along the end-effector z -axis, so the selected tool point then sits on that axis directly beneath the TCP.

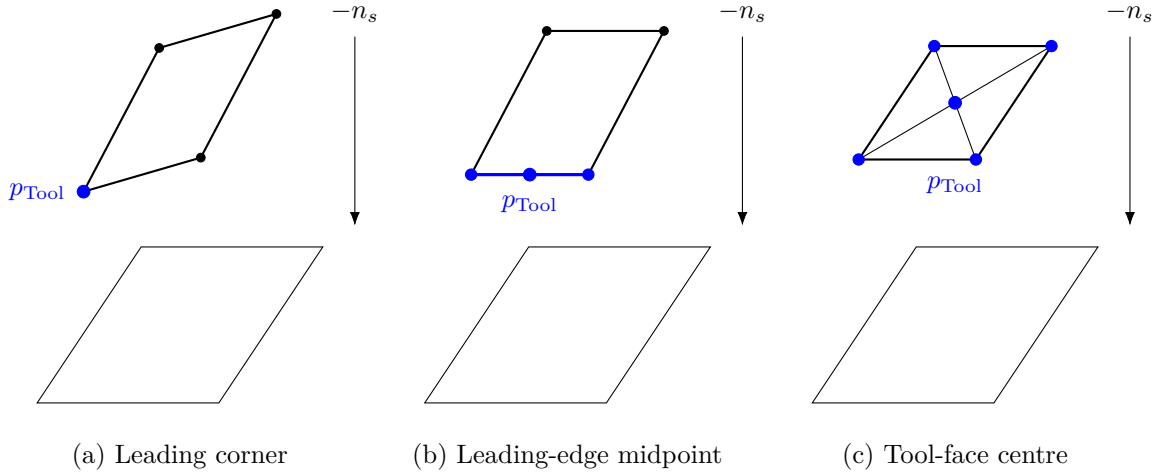


Figure 3.4: Principal selected tool points for the rectangular tool face: leading corner, leading-edge midpoint, and tool-face centre.

The minimum clearance h_{Tool} determines when approach ends. The offset $r_{\text{Tool,EE}}$ determines which physical tool point the set-up reference uses. At the clearance transition the selected offset is retained in end-effector coordinates and then held constant throughout set-up.

Three points remain distinct in this construction. The impedance law regulates the Cartesian reference point p_{TCP} . The selected tool point p_{Tool} belongs to the physical tool ge-

ometry, and its base-frame offset $r_{\text{Tool}} = R_{\text{EE}} r_{\text{Tool,EE}} = p_{\text{Tool}} - p_{\text{TCP}}$ changes with the tool orientation. The virtual CoC p_c of Section 2.7 is a software-defined impedance reference and carries no tool geometry.

With the optional pre-set-up gate enabled, the captured pose is held until operator confirmation, and the set-up references are then captured from the current measured state.

3.2.4 Contact Set-Up

In the reported surface-contact runs, set-up begins directly at the clearance transition. Its Cartesian reference is constructed from the selected tool point and end-effector pose captured at that instant. The captured orientation $R_{\text{EE,clearance}}$ and TCP position $p_{\text{TCP,clearance}}$ are taken at the clearance-transition instant $t_{\text{set},0}$,

$$R_{\text{EE,clearance}} = R_{\text{EE}}(t_{\text{set},0}), \quad p_{\text{TCP,clearance}} = p_{\text{TCP}}(t_{\text{set},0}). \quad (3.16)$$

The corresponding selected tool-point position $p_{\text{Tool,clearance}}$ is

$$p_{\text{Tool,clearance}} = p_{\text{TCP,clearance}} + R_{\text{EE,clearance}} r_{\text{Tool,EE}}. \quad (3.17)$$

Its orthogonal projection onto the calibrated surface plane defines the surface reference point $p_{\text{Tool},0}$,

$$p_{\text{Tool},0} = p_{\text{Tool,clearance}} - n_s n_s^\top (p_{\text{Tool,clearance}} - p_s). \quad (3.18)$$

$p_{\text{Tool},0}$ is therefore the point on the calibrated plane directly below the selected tool point at the clearance transition.

Set-up generates the contact press evaluated in the experiments. The scalar coordinate $s_{\text{set}}(t)$ gives the selected tool point's commanded displacement from $p_{\text{Tool},0}$ towards the surface. Its initial value is the negative signed surface coordinate of that point,

$$s_{\text{set}}(t_{\text{set},0}) = -n_s^\top (p_{\text{Tool,clearance}} - p_s). \quad (3.19)$$

The selected point remains above the surface at the transition, so $s_{\text{set}}(t_{\text{set},0}) < 0$. This

initial value makes the first set-up reference equal to $p_{\text{Tool,clearance}}$ and avoids a position step.

The coordinate advances at the configured press speed v_{set} from $t_{\text{set},0}$ until it reaches the configured endpoint $s_{\text{set,max}}$, then remains constant:

$$s_{\text{set}}(t) = \begin{cases} s_{\text{set}}(t_{\text{set},0}) + v_{\text{set}}(t - t_{\text{set},0}), & t_{\text{set},0} \leq t < t_{\text{set,max}}, \\ s_{\text{set,max}}, & t \geq t_{\text{set,max}}. \end{cases} \quad (3.20)$$

The desired selected-tool-point position $p_{\text{Tool},d}(t)$ is

$$p_{\text{Tool},d}(t) = p_{\text{Tool},0} - s_{\text{set}}(t)n_s. \quad (3.21)$$

Appendix C records the configured press speed and endpoint.

Throughout set-up the desired orientation is

$$R_d(t) = R_{\text{EE,clearance}}. \quad (3.22)$$

The set-up trajectory is defined for the selected physical tool point, whereas the Cartesian impedance law regulates the TCP. The desired tool-point position must therefore be converted to the corresponding TCP reference. At the retained orientation, their rigid-body relation is

$$p_{\text{Tool},d}(t) = p_d(t) + R_{\text{EE,clearance}}r_{\text{Tool,EE}}. \quad (3.23)$$

Solving this relation gives the desired TCP position $p_d(t)$ as

$$p_d(t) = p_{\text{Tool},d}(t) - R_{\text{EE,clearance}}r_{\text{Tool,EE}}. \quad (3.24)$$

During the linear part of the ramp, the desired TCP velocity is $\dot{p}_d = -v_{\text{set}}n_s$. It is zero after $s_{\text{set,max}}$ is reached. The controller generates the press by prescribing the selected tool-point trajectory and reconstructing the TCP target that realises it. Because the surface restricts this commanded motion, the resulting Cartesian position error generates the press. The configured endpoint lies below the surface and therefore defines a spring

reference rather than an expected physical travel.

Figure 3.5 shows the complete construction from the projected tool point to the TCP target.

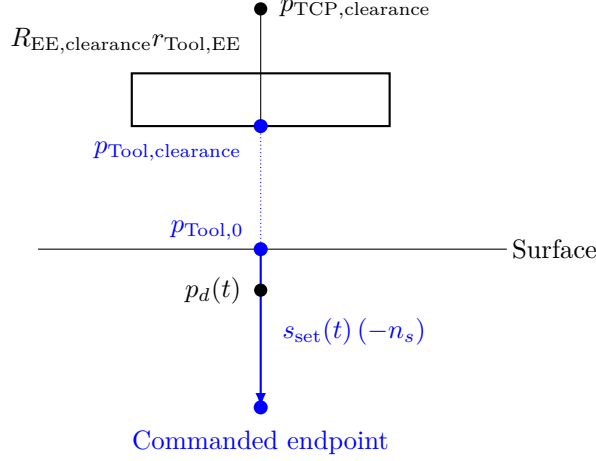


Figure 3.5: Set-up reference generation for the selected tool point.

No time-varying tipping trajectory is commanded during set-up. The held orientation reference acts as a finite-stiffness rotational spring rather than as a rigid pose constraint, so finite rotational compliance permits contact-induced end-effector rotation. When the virtual CoC is enabled, the point-shifted matrices of Section 3.2.4 additionally couple the translational press to the commanded moment; the same set-up position reference is used in both cases.

The model-estimated external wrench supplied by libfranka is stored at the clearance transition for the optional termination condition. After the configured minimum duration, set-up can terminate when the estimated external moment change exceeds the configured threshold, or through the independent timeout. Every reported contact run terminated through the timeout, so that threshold did not affect the reported results. With the pre-grinding gate active, the controller remains in the set-up branch while waiting for operator confirmation, and the coupled impedance stays active where it was selected.

The impedance acting on that reference changes at the same transition, from the approach set to the contact set-up set. Three choices determine it: the directional stiffness, the damping associated with those directions, and whether the impedance reference point is the TCP or a shifted virtual CoC. The controller constructs the directional gains first and applies the compliance-centre transformation afterwards.

Directional Cartesian stiffness and damping are defined relative to the surface frame and transformed to the robot base frame by the congruence of Section 2.6. Each phase of the sequence activates its own gain set in the calibrated surface frame, and the set-up translational frame remains configurable. Set-up and grinding draw on the same source stiffness and damping blocks; the phase transition changes the generated Cartesian reference and the choice between the coupled and decoupled branch. A phase gate can temporarily replace the active matrices with dedicated hold gains while the current reference is maintained. The numerical gains used in the experiments are given in Chapter 4.

For phase groups configured to use inertia-scaled damping, each directional damping coefficient follows from the paired stiffness entry k_i and the directional Cartesian inertia $\lambda_i(q)$ of the same translational or rotational direction:

$$d_i(q) = 2\zeta\sqrt{\lambda_i(q)k_i}, \quad (3.25)$$

where ζ is the configured directional damping factor. For a single decoupled direction with $\zeta = 1$, this is the critical-damping coefficient of that local second-order model.

The directional inertia is evaluated from the current joint-space mass matrix $M(q)$ and Jacobian $J(q)$ supplied by the libfranka model, and then resolved along the axes of the active gain block. The inverse $M(q)^{-1}$ is never formed: an LDLT factorisation of $M(q)$ solves the corresponding linear system instead, and a small regularisation term keeps the subsequent inversion conditioned. The result is cached by phase group and recalculated when that group is entered again, so the current robot configuration is reflected in the damping; configured coefficient bounds, validity checks, and the fallback damping matrix are recorded in Appendix C.

The reported surface-contact sequence used inertia-scaled damping in the phase groups stated in Chapter 4.

The point-shift formulation derived in Section 2.7 is implemented with the lever $r_{c,0} = p_c - p_{\text{TCP}}$. Two definitions are supported, because the meaning of a fixed displacement depends on the frame that holds it.

A tool-fixed displacement is configured in end-effector coordinates. Its base-frame vector $r_{c,0} = R_{\text{EE}}r_{c,\text{EE}}$ is therefore updated from the measured end-effector orientation in every

callback and rotates as the end effector rotates. A surface-fixed displacement is configured in surface coordinates, so $r_{c,0} = R_{\text{surface}} r_{c,S}$ stays constant during set-up, the surface frame being fixed for one controller session. The configuration loader requires exactly one of the two definitions to be active when the virtual centre is enabled, and rejects a parameter set selecting both or neither. The two definitions are compared experimentally in Appendix D.

The source translational and rotational matrices are assembled into block-diagonal spatial gains at p_c , and the same congruence is applied independently to stiffness and damping:

$$K_{\text{TCP}} = \text{Ad}^\top(r_{c,0}) K_c \text{Ad}(r_{c,0}), \quad D_{\text{TCP}} = \text{Ad}^\top(r_{c,0}) D_c \text{Ad}(r_{c,0}). \quad (3.26)$$

For a tool-fixed displacement the shifted base-frame matrices are updated whenever the measured end-effector orientation changes, whereas a surface-fixed displacement leaves them constant as long as the source gains do not change. Setting $r_{c,0} = 0$ makes the adjoint the identity and returns the decoupled block-diagonal form.

During set-up this point-shifted impedance allows the commanded normal press to contribute to the rotational response. The influence of the compliance-centre position on that response is evaluated experimentally in Chapter 5. The coupled branch is used only in the controller modes configured for a virtual CoC; the cases that used it are specified in Chapter 4 and Appendix C.

3.2.5 Grinding

After set-up, the grinding phase retains the normal press and superimposes a smooth tangential motion using the decoupled impedance. No reported run entered this phase; the experimental evaluation therefore ends at the pre-grinding gate.

3.3 Cartesian Pose Hold

Cartesian pose hold captures the current end-effector pose at its entry instant t_0 and uses the fixed references $p_d = p_{\text{EE}}(t_0)$ and $R_d = R_{\text{EE}}(t_0)$. Its dedicated hold impedance is expressed in the base frame. Surface approach, clearance detection and set-up reference

generation take no part in it, and its reported trials used fixed hold damping rather than the inertia-scaled rule of Section 3.2.4.

Only the secondary torque varies inside this configuration. The four selectable null-space configurations are shown in Figure 3.6: no null-space torque, singular-value conditioning, projected damping, and the two terms together. The Cartesian reference and Cartesian impedance are unchanged between them.

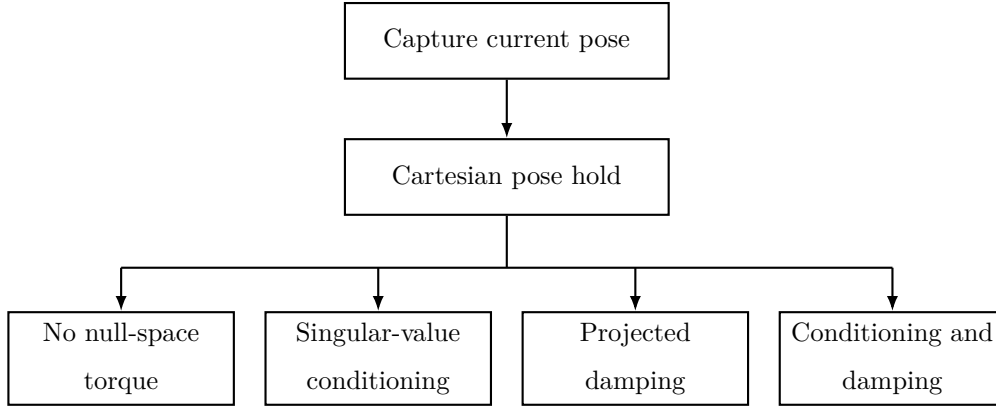


Figure 3.6: Cartesian pose hold with four selectable null-space configurations.

The pose-hold study of Section 4.6 compares four settings under one internally commanded disturbance. These settings are no null-space torque, projected damping, and singular-value conditioning at two commanded magnitudes. They do not correspond to the four modes one for one. The combined mode runs in the surface-contact sequence and was not compared in pose hold. The disturbance, its timing and the evaluated quantities belong to Section 4.6.

The implementation adds two choices not fixed by Section 2.8. The pseudoinverse threshold is the configured relative tolerance multiplied by the largest current singular value. The resulting joint-torque projector is symmetrised numerically, and the same matrix produces the recorded projected joint velocity $\dot{q}_{\text{null}}$. For singular-value conditioning, the controller evaluates the Jacobian at $q \pm \alpha_{\text{probe}} v_7$ and compares the smallest singular value at each probe. This comparison uses the current robot transformations, and no conditioning torque is requested inside the configured deadband. The surface-contact sequence holds one combined setting throughout, so the secondary controller is not a variable in those comparisons. Every numerical value is given in Chapter 4 and Appendix C.

3.4 Robot-Side Safety and Command Handling

The joint-torque command of Equation (3.1) is returned directly to libfranka. No application-side saturation or torque-rate limiter is applied after the Cartesian, null-space and Coriolis terms have been assembled, so the configured collision and reflex monitoring is the independent protection layer during the reported experiments. Its joint-torque and Cartesian-wrench thresholds are recorded in Section 4.1.3. A robot communication, motion-generation or torque-control error leaves the callback through `franka::Exception` and ends the control session.

Everything the robot needs before torque control is completed before it starts: the connection, error recovery, the collision configuration, the configuration validation, the gripper action and the logging-buffer allocation. A libfranka joint-motion generator then brings the robot to a configured initial joint posture, or the run starts from a pose captured after manual guidance, with the posture used in the reported experiments recorded in Appendix C. Operator input and file writing are handled outside the real-time thread, so neither adds a blocking operation to the joint-torque callback.

3.5 Real-Time Data Recording

Experimental signals are stored during torque control in a preallocated in-memory ring buffer, so the callback performs no file access and grows no buffer dynamically. Each sampled row carries the measured state, the active reference, the calculated error and wrench, the controller phase, the secondary-controller state, and the returned joint torque. When the allocated capacity is reached the write index wraps and new rows replace the oldest, which keeps the memory bounded while control continues. After the callback ends, the retained samples are written to the CSV file in chronological order.

Values captured at the clearance transition refer to the geometric event of Section 3.2.3. The complete column-level description is given in Appendix B, and the experimental metrics calculated from these signals are defined in Chapter 4.

Chapter 4 next specifies the physical system, the calibration procedure, the experimental parameter matrix, the contact procedure, and the evaluation quantities used to assess this controller.

Chapter 4

Experimental Methodology

This chapter describes how the controller developed in Chapter 3 was evaluated experimentally. The methodology is organised in the order in which the experiments can be reproduced: the physical system is introduced first, followed by the surface reference and tool calibration, the common contact-test configuration and procedure, the main surface-contact matrix, the quantities used for evaluation, and the separate Cartesian pose-hold study for the null-space controller. Supporting checks of the compliance-centre geometry are reported in Appendix D.

The main data set consists of surface-contact measurements in which the commanded tool orientation offset, directional Cartesian stiffness, and virtual CoC p_c were varied under otherwise matched conditions. The primary controller-response quantity is the signed set-up response. The TCP-height flatness classification is retained as an appendix check, while the commanded wrench supports the Case-D mechanism plot. A second, independent experiment evaluates the null-space terms under a repeatable internally commanded disturbance.

4.1 Experimental System

4.1.1 Robot, Tool, and Surface

The experiments were performed with a seven-degree-of-freedom Franka Emika Panda operated through the Franka Control Interface. Joint-torque commands were transmitted

through libfranka at the nominal control rate of 1 kHz, corresponding to a sampling period of 1 ms. The real-time controller, Cartesian impedance law, contact geometry, and phase sequence are described in Chapter 3.

The rectangular grinding tool was held by the Franka Hand using custom gripper fingers. The tool face measured 40×120 mm, with the width along X_{EE} and the length along Y_{EE} . Its centre was located 20 mm along $+Z_{EE}$ from the reported end-effector origin. The tool was clamped by custom pads screwed to the gripper fingers. Mechanical clearance in that clamping allowed the tool to rotate relative to the gripper by approximately $\pm 2^\circ$ about y_{EE} , which corresponds approximately to the t_2 direction in the flat experimental configuration. The robot does not measure this relative rotation: it is absent from the joint and end-effector state. The $\pm 2^\circ$ was measured in the unloaded condition, and the instantaneous relative tool–gripper rotation was not tracked during the contact experiments. It is therefore a mechanical property of the mounting rather than a calibrated measurement uncertainty.

Orientation quantities reconstructed from R_{EE} and the calibrated tool normal consequently describe the tool orientation under the assumption of a fixed tool-to-end-effector transformation, and any motion of the tool within the gripper is not included in them.

The surface was a plane plate mounted on a raised base board. All reported measurements used dry contact. The physical surface orientation and the tool normal were calibrated separately before the measurements, as described in Section 4.2.

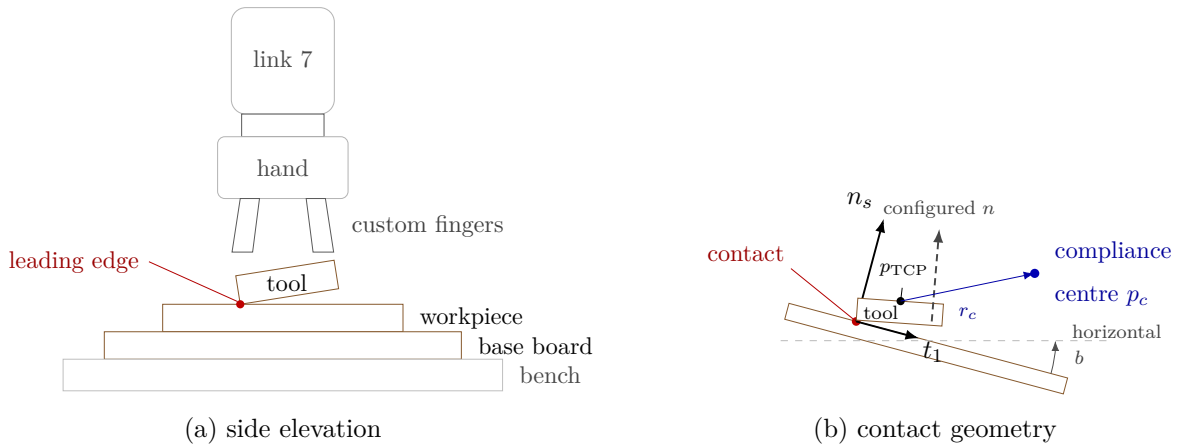


Figure 4.1: Experimental system and surface-frame contact geometry.

4.1.2 Real-Time Computer and Software

The controller computer ran Ubuntu 20.04.3 with the 5.9.1-rt20 Linux kernel and the PREEMPT_RT patch. The controller was compiled using g++ 9.3, C++14, and optimisation level -O2. Eigen 3.3.7 was used for matrix calculations and libfranka 0.7.1 for communication with the robot.

Before the measurements, the computer-robot connection was checked with the `communication_test` program supplied with libfranka 0.7.1 [6]. The program reports the fraction of control commands received by the robot and issues its lower communication-quality warning below an average command-success rate of 0.90. The experimental system passed this criterion. This check verifies communication quality for operation through the nominal 1 kHz interface; it is not a worst-case execution-time or scheduling-jitter measurement.

4.1.3 Operating and Safety Configuration

Before torque control started, the robot-side collision behaviour was configured with equal acceleration and nominal joint-torque thresholds of 140 N m. The six Cartesian force and moment channels used thresholds of 240 N or 240 N m, according to the component type. These thresholds supplied the Franka reflex mechanism and were kept unchanged throughout the reported measurements.

The complete run-specific controller configuration is given in Appendix C.

4.2 Surface Reference and Tool Calibration

Two geometric quantities were required before the contact experiments could be performed: a surface reference expressed in the robot base frame, and the normal direction of the physical tool face expressed in the end-effector frame. Each came from its own calibration procedure.

Figure 4.2 sets out both procedures. The surface-reference construction of Section 4.2.1 supplies the surface reference p_s , n_s and $R_{\text{surface}} = [t_1 \ t_2 \ n_s]$ in the base frame, and the tool-normal calibration of Section 4.2.2 supplies $n_{\text{Tool,EE}}$ in end-effector coordinates. The

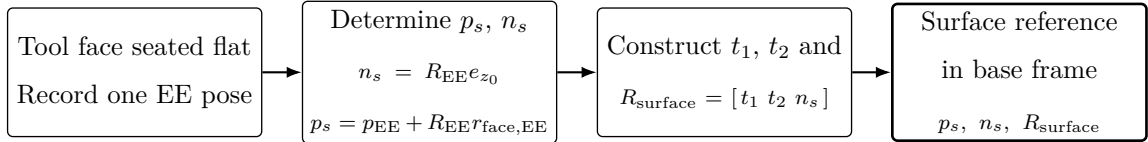
second calibration is needed because the physical face is not exactly perpendicular to the nominal $+Z_{EE}$ axis.

Each symbol carries the frame it is expressed in. $n_{Tool,EE}$ is the tool-face normal in end-effector coordinates, and $n_{Tool,0}$ is the same physical direction in the base frame. Only the first is calibrated. The second follows from it during operation, as Section 3.2.1 sets out. It is used only for the pose-based consistency check reported in Appendix D.7.

Neither calibration entered the other. The tool normal calibrated afterwards was not used to revise the surface reference, and determining $n_{Tool,EE}$ used the relative orientations of the seated yaw poses alone, with no value of n_s . That separation matters, because the calibrated $n_{Tool,EE}$ later differed from nominal $+Z_{EE}$ by approximately 1.56° .

Consequently the surface frame used by the controller is a configured geometric reference rather than an independent measurement of the physical plane normal. It was taken from the nominal end-effector axis at one seated pose. The 1.56° is the difference between those two orientation references, and it applies equally to every reported run. Since it limits an absolute interpretation of the physical tool–surface angle, the main comparisons report the signed set-up response of Section 4.5.1.

Surface-reference construction



Tool-normal calibration

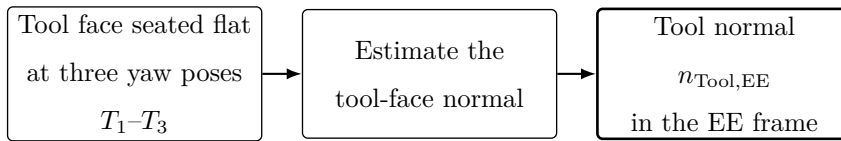


Figure 4.2: Surface-reference construction and tool-normal calibration.

4.2.1 Surface-Reference Construction

The complete tool face was seated on the plane and one end-effector pose was recorded. Its nominal $+Z_{EE}$ direction supplied the surface normal n_s , and the tool-face centre supplied the plane point p_s . Projecting the base-frame x direction onto the plane gave t_1 ; $t_2 = n_s \times t_1$ completed $R_{surface} = [t_1 \ t_2 \ n_s]$. The stored surface tilts were -1.585° about

base x and $+0.988^\circ$ about base y . The calibrated values and their construction are collected in Appendix C.

4.2.2 Tool Normal Calibration

The complete tool face was seated at three yaw orientations. The direction in end-effector coordinates that remained most nearly invariant in the base frame was fitted from those three captures. The resulting tool normal $n_{\text{Tool,EE}}$ differed from nominal $+Z_{\text{EE}}$ by approximately 1.56° . Its value and the minimisation used for the fit are given in Appendix C.

4.2.3 Calibrated Experimental Geometry

The rectangular face geometry used by the controller is listed in Table 4.1. The selected-tool-point construction and the reconstruction of the TCP target are described in Section 3.2.3 and Section 3.2.4, respectively.

Table 4.1: Calibrated tool geometry.

Quantity	Value in end-effector coordinates
Tool-face centre	$[0, 0, 0.020]^\top \text{ m}$
Half-width	$[0.020, 0, 0]^\top \text{ m}$
Half-length	$[0, 0.060, 0]^\top \text{ m}$
Tool-point selection tolerance, ε_{sel}	0.1 mm

4.3 Common Contact-Test Configuration and Procedure

All surface-contact cases used the same calibrated geometry, phase sequence, normal press trajectory, fixed gain entries, and null-space configuration except where a parameter was explicitly varied. This section defines that common experimental basis before the individual cases are introduced in Section 4.4.

4.3.1 Common Cartesian Gains and Damping

The phase-dependent gains are diagonal in the calibrated surface frame $[t_1, t_2, n_s]$ before any compliance-centre shift. Table 4.2 gives the entries used by every reported contact run, together with the damping ratio ζ of the corresponding phase. The pre-set-up gate was configured with the same stiffness as set-up and was disabled in the reported runs, so it did not act as a separate phase.

Table 4.2: Common Cartesian gains and damping ratio by phase.

Phase	K_p [N/m]			K_R [N m/rad]			ζ
	t_1	t_2	n_s	t_1	t_2	n_s	
Orientation and approach	150	150	150	180	180	90	1.9
Set-up	2000	2000	350	5	5	50	1.0

During orientation and approach the translational stiffness was isotropic and the rotational stiffness was higher about the surface tangents. In set-up the lower normal entry limits the elastic normal-force increase for a given normal position error, while the higher tangential entries restrict motion in the surface plane. The low rotational stiffnesses about t_1 and t_2 permit contact-induced alignment rotation, and the higher entry about n_s mainly resists in-plane spin of the tool face. Because the set-up translational stiffness $K_{p,\text{set}}$ is diagonal in the surface frame, the effective normal stiffness $K_{p,n} = n_s^\top K_{p,\text{set}} n_s$ equals the tabulated normal entry of 350 N/m.

The directional damping coefficients were calculated from the Cartesian inertia and the active stiffness at the tabulated damping ratio, as described in Section 3.2.4. In contrast, the fallback damping values, used only if the inertia-based calculation could not be applied, are provided in Appendix C.

4.3.2 Contact Sequence and Set-Up Motion

The common trajectory and transition parameters are collected in Table 4.3. These settings were used throughout the surface-contact experiments unless explicitly stated otherwise.

Table 4.3: Common phase-transition and trajectory parameters for the surface-contact experiments.

Quantity	Value	Role
Minimum orientation time	1.0 s	Prevents an immediate phase transition.
Tool-axis transition tolerance	0.016 rad	Maximum remaining tool-axis error before descent.
Spin transition tolerance	0.009 rad	Maximum remaining normal-axis spin error.
Orientation timeout / maximum rate	8.0 s / 20°/s	Bounds the orientation phase.
Descent speed	0.1 m/s	Reference speed along the descent direction.
Maximum descent distance	0.640 m	Terminates an unsuccessful approach.
Tool clearance, $h_{\text{clearance}}$	0.020 m	Minimum tool-face clearance at the transition to set-up.
Minimum set-up time	2.0 s	Earliest activation of the moment-based exit condition.
Set-up timeout	5.0 s	Time-based set-up termination.
Moment-change threshold	150 N m	Alternative set-up termination based on the model-estimated external moment.
Set-up push speed	0.080 m/s	Rate of the signed press coordinate.
Set-up push endpoint	+0.240 m	Final signed Cartesian spring reference.
Pre-set-up gate	disabled	Approach transfers directly to set-up.

Table 4.3: Common phase-transition and trajectory parameters for the surface-contact experiments (continued).

Quantity	Value	Role
Pre-grind gate	enabled	Holds the reached pose at the end of the reported set-up sequence.

Before each run, the active parameter set was loaded and the robot was moved to the configured initial joint posture. The calibrated geometry, the phase gains and damping, and the compliance-centre definition were prepared before the torque callback started.

Approach ended when the part of the rectangular tool face closest to the calibrated plane reached the tool-clearance threshold in Table 4.3. This transition was defined geometrically rather than by a force threshold. At the same instant, the controller stored the TCP pose, the orientation reference, the selected tool point, and its projection onto the plane. The model-estimated external wrench was additionally stored for the optional wrench-change termination condition described in Section 3.2.4.

The set-up press endpoint of +0.240 m is a Cartesian spring reference beyond the calibrated plane and is not interpreted as a physical penetration of the same magnitude. The reference reached it after approximately 3.25 s and was then held. Throughout set-up the rotational reference remained the orientation captured at the clearance transition, so the measured set-up response results from the contact interaction rather than from tracking a time-varying orientation command. The TCP-centred and stiffness-variation conditions used the decoupled Cartesian impedance during set-up. Conditions with a displaced CoC used the point-shifted 6×6 impedance associated with the selected compliance-centre position.

Every reported contact run ended at the 5 s set-up timeout, and the moment-change threshold was reached in none of them. The enabled pre-grind gate then held the reached pose, so the recorded data cover orientation, approach, and set-up only.

Fixed null-space configuration. Projected null-space damping and singular-value conditioning were active together throughout the surface-contact measurements. Keeping this secondary controller fixed prevents changes in the null-space policy from be-

coming an additional variable across the surface-contact experiments. The gains were $d_{\text{null}} = 2 \text{ N m s/rad}$ and $k_{\sigma} = 1 \text{ N m}$.

This is the combined secondary-torque mode of the implementation described in Section 3.3. The contact measurements therefore evaluate the surface-alignment controller with both null-space terms active. Their individual effects are examined separately in the pose-hold experiment of Section 4.6.

4.3.3 Repetitions and Data Recording

The recorded signals and CSV structure are documented in Section 3.5 and appendix B. Three repetitions were recorded for each parameter setting. Results are reported as arithmetic means, with sample standard deviations given for the main comparisons and where relevant in the supporting checks. The main surface-contact study comprises 111 runs across 37 settings. A further 54 runs across 18 settings were performed as supporting checks and are reported in Appendix D. In total, 165 surface-contact runs across 55 settings were recorded. No run was discarded. Together with the 12 pose-hold runs of Section 4.6, the complete experimental data set contains 177 recorded runs.

Within each controlled comparison, the calibrated geometry, common phase parameters, fixed gains, and null-space settings remained unchanged. Only the parameter assigned to the corresponding comparison was varied.

4.4 Main Surface-Contact Experimental Matrix

The main surface-contact study separates the effects of the principal controller parameters. Case A establishes the TCP-centred reference, where $r_c = 0$. In Cases B and C, rotational and translational stiffness were varied independently so that each gain could be evaluated without changing the other simultaneously. In Case D the tangential compliance-centre position was varied across both principal surface tangents and both signs of the commanded orientation offset.

Unless stated otherwise, the non-zero compliance-centre displacements of the main study were configured tool-fixed, in end-effector coordinates. The tangential coordinates reported for Case D are the surface-frame directions those settings realise in the flat tar-

get orientation. The definition frame itself is examined in the supporting check of Appendix D.3.

The orientation-offset-magnitude comparison, the CoC definition frame, displacement along the tool axis, and intermediate tangent-plane directions are treated as supporting checks in Appendix D.

Table 4.4 summarises the parameter variations of the main study. Each listed setting was repeated three times unless it is a reference condition counted with an earlier case.

Table 4.4: Main surface-contact cases and parameter variations.

Case	Varied quantity	Commanded tool orientation offset	Tested values	Runs	Evaluated effect
A	CoC at the TCP	None and $\pm 10^\circ$ about t_1 or t_2	$r_c = 0$.	15	Contact response without CoC coupling.
B	Rotational stiffness	$+10^\circ$ about t_1 or t_2	Command-axis $K_R = 15$ and 50 N m/rad .	12	Influence of rotational stiffness.
C	Translational stiffness	$+10^\circ$ about t_1 or t_2	Cross-axis $K_p = 300$ and 800 N/m .	12	Influence of translational stiffness.
D	Tangential CoC position	$\pm 10^\circ$ about t_1 or t_2	$-40, -20, -10, +10, +20, +40 \text{ mm}$ along the tangent perpendicular to the commanded rotation.	72	Direction and sign dependence.

The run counts in Table 4.4 give the three scheduled repetitions of each setting. A condition that a case shares with an earlier one is counted with that earlier case.

In Cases B and C, only one Cartesian stiffness entry was varied within each comparison. The remaining gain entries were kept at the common values of Table 4.2.

4.5 Evaluation Quantities

The primary response quantity for the surface-contact experiments is the signed set-up response γ_{t_i} . It is calculated from the measured end-effector orientation at the end of set-up and the orientation reference held through that phase. The commanded Cartesian

wrench is retained for the Case-D mechanism plot.

4.5.1 Signed Set-Up Response

Let $t_{\text{set,start}}$ and $t_{\text{set,end}}$ denote the measurement instants at the beginning and end of set-up. The corresponding measured end-effector orientations relative to the robot base frame are

$$R_{\text{set,start}} = R_{\text{EE}}(t_{\text{set,start}}), \quad R_{\text{set,end}} = R_{\text{EE}}(t_{\text{set,end}}). \quad (4.1)$$

Set-up holds the orientation captured at the clearance transition as its rotational reference, and the measured orientation coincides with it at the start of the phase, so $R_{\text{set,start}}$ is that held reference. The relative rotation from the measured end orientation back to it is therefore

$$R_{\text{set}} = R_{\text{set,start}} R_{\text{set,end}}^\top. \quad (4.2)$$

Writing this relative rotation in axis-angle form as $R_{\text{set}} = R(u_{\text{set}}, \phi_{\text{set}})$ gives the measured set-up rotation vector γ_0 in base coordinates:

$$\gamma_0 = \phi_{\text{set}} u_{\text{set}} = e_R \Big|_{\text{end of set-up}}. \quad (4.3)$$

This quantity is obtained from the measured end-effector orientation alone, and the calibrated tool normal does not enter it. It is therefore independent of the relative rotation between the tool and the gripper described in Section 4.1.1, which was not tracked during the contact experiments. The surface frame enters solely as the directions the rotation is resolved along:

$$\gamma_S = R_{\text{surface}}^\top \gamma_0 = \begin{bmatrix} \gamma_{t_1} & \gamma_{t_2} & \gamma_n \end{bmatrix}^\top, \quad \gamma_{t_i} = t_i^\top \gamma_0, \quad i \in \{1, 2\}. \quad (4.4)$$

The reported quantity γ_{t_i} is the signed set-up response about surface tangent t_i . It represents the end-of-set-up orientation change relative to the held set-up reference, resolved along that tangent. Its sign is chosen such that a response reducing a commanded offset

carries the same sign as that offset: a correcting response is positive for a positive offset and negative for a negative one, and its magnitude is the size of the correction. Reading the sign alone is therefore not enough, since the sign of the commanded offset fixes which direction counts as correcting. The measurements confirm the correspondence: for the $+10^\circ$ command about t_1 at the TCP, $\gamma_{t_1} = +7.57^\circ$ accompanies a fall in the pose-based alignment angle from 9.32° to 3.38° , whereas the $+10^\circ$ command about t_2 gives $\gamma_{t_2} = -1.63^\circ$ with that angle rising from 10.07° to 11.81° . This quantity is used for all main comparisons in Chapter 5.

4.5.2 Supporting Geometric Checks

A separate pose-based alignment angle calculated from the calibrated tool normal is retained only as a supporting consistency check in Appendix D.7. It is not used for the main comparisons because it is unsigned and depends on an assumed fixed relation between the tool and end effector.

A separate geometry-based flatness classification is obtained from the final TCP height relative to the calibrated plane and the height expected from the calibrated rectangular tool geometry when the tool face is flat. This classification uses the end-effector position and calibrated geometry; it is not a direct angular measurement of the physical tool face. The TCP-height classification is therefore reported separately from the primary end-effector set-up rotation.

4.5.3 Commanded Wrench Quantities

The commanded normal force and commanded TCP moment used in the controller-response plots are obtained from $F = [f^\top, m^\top]^\top$:

$$F_n = n_s^\top f, \quad M_{t_i} = t_i^\top m. \quad (4.5)$$

F_n is signed, and it is negative while the tool presses into the surface. Both quantities are controller commands, and the plotted moment is therefore the complete TCP moment generated by the point-shifted spring-damper law.

4.6 Secondary Null-Space Pose-Hold Experiment

The surface-contact measurements keep projected damping and singular-value conditioning active together. A separate Cartesian pose-hold experiment was therefore used to compare their individual effects under a repeatable internally commanded disturbance.

4.6.1 Experimental Configuration

The pose-hold controller maintained the initial TCP pose with the gains of Table 4.5. Every entry was isotropic, so each matrix is the stated value times the identity. Unlike the contact-sequence phases, the twelve pose-hold trials used fixed Cartesian damping rather than inertia-scaled damping.

Table 4.5: Isotropic Cartesian gains of the pose-hold trials.

Quantity		Value
Translational stiffness	$K_{p,\text{hold}}$	2000 N/m
Rotational stiffness	$K_{R,\text{hold}}$	50 N m/rad
Translational damping	$D_{p,\text{hold}}$	50 N s/m
Rotational damping	$D_{R,\text{hold}}$	8 N m s/rad

Four null-space settings were tested with three repetitions each: no null-space torque, projected damping with $d_{\text{null}} = 2 \text{ N m s/rad}$, sigma conditioning with $k_{\sigma} = 1.5 \text{ N m}$, and sigma conditioning with $k_{\sigma} = 2.0 \text{ N m}$. The probe step, sigma deadband, and relative SVD threshold remained at 0.08 rad, 10^{-6} , and 10^{-4} , respectively.

4.6.2 Commanded Disturbance

The disturbance represents the joint-torque equivalent of a commanded point force acting at a specified location on link 3. The point was defined by the link-frame offset

$$r_p = \begin{bmatrix} 0 \\ 0 \\ 0.200 \end{bmatrix} \text{ m.} \quad (4.6)$$

At the initial posture q_{init} , the structural null direction $v_7(q_{\text{init}})$ and the translational point Jacobian $J_p(q_{\text{init}})$ define the fixed base-frame force direction

$$u_f = \frac{J_p(q_{\text{init}})v_7(q_{\text{init}})}{\|J_p(q_{\text{init}})v_7(q_{\text{init}})\|_2}. \quad (4.7)$$

The commanded force magnitude was $F_d = 20$ N. The virtual force and its equivalent joint torque were

$$f_d(t) = F_d s_d(t) u_f, \quad \tau_{\text{dist}}(t) = J_p(q(t))^{\top} f_d(t). \quad (4.8)$$

Here u_f is the fixed unit force direction defined in Equation (4.7), and $s_d(t) \in [0, 1]$ is a dimensionless time-scaling function that switches the commanded virtual disturbance smoothly on and off. It was defined as

$$s_d(t) = \begin{cases} 0, & t < 5 \text{ s}, \\ \frac{1}{2} \left[1 - \cos\left(\pi \frac{t - 5 \text{ s}}{2 \text{ s}}\right) \right], & 5 \text{ s} \leq t < 7 \text{ s}, \\ 1, & 7 \text{ s} \leq t < 8 \text{ s}, \\ \frac{1}{2} \left[1 + \cos\left(\pi \frac{t - 8 \text{ s}}{1 \text{ s}}\right) \right], & 8 \text{ s} \leq t < 9 \text{ s}, \\ 0, & t \geq 9 \text{ s}. \end{cases} \quad (4.9)$$

Thus the disturbance increased smoothly from zero to the full commanded magnitude between 5 and 7 s, remained at full magnitude until 8 s, and decreased smoothly to zero by 9 s. The equivalent disturbance torque was limited to

$$\|\tau_{\text{dist}}\|_2 \leq 3 \text{ N m}.$$

The disturbance is internal to the controller and therefore provides a repeatable comparison input; it is not a separately applied physical force.

The equivalent torque is formed from the point Jacobian alone and carries no null-space projector, so it is a joint-torque excitation rather than a null-space command. Its image therefore has a component along the six Cartesian directions as well as along the redundant

direction, and the Cartesian impedance resists that component while the disturbance acts. Equation (4.10) therefore checks that the commanded pose was retained rather than assuming it.

In every trial the selected null-space law was active from the start of the pose-hold phase. The first 5 s therefore formed a pre-disturbance settling interval, during which the conditioning torque of the sigma-only settings could already move the redundant configuration. The disturbance then acted from 5 to 9 s, followed by an observation interval until the end of the trial. No secondary term was enabled or disabled within a trial: the sigma-conditioning torque remained active throughout the complete sigma-only trial and was not switched on after the disturbance had been removed.

4.6.3 Evaluation Quantities

All quantities in this subsection are evaluated over the interval in which the internally commanded virtual disturbance acts, $t \in [5, 9]$ s. The first criterion checks whether the primary Cartesian pose-hold objective is maintained while the disturbance and the selected null-space law act. The predefined position-retention criterion is

$$\max_{5\text{ s} \leq t \leq 9\text{ s}} \|e_p(t)\|_2 < 2 \text{ mm}, \quad (4.10)$$

where $e_p(t)$ is the Cartesian position-error vector. A trial therefore satisfies the criterion when the TCP position-error norm remains below 2 mm throughout the commanded disturbance interval.

The redundant joint motion is evaluated from the projected joint velocity

$$\dot{q}_{\text{null}}(t) = N_q(q(t))\dot{q}(t), \quad (4.11)$$

where $N_q(q)$ is the kinematic null-space projector. The vector $\dot{q}_{\text{null}} \in \mathbb{R}^7$ has units of rad/s and represents the component of the measured joint velocity that lies in the instantaneous kinematic null space.

Two complementary quantities are formed from this projected velocity. The first is the cumulative projected null-space motion

$$E_N = \int_{5s}^{9s} \|\dot{q}_{\text{null}}(t)\|_2 dt. \quad (4.12)$$

Because the norm is taken before integration, motion in opposite directions contributes positively in both cases. E_N therefore measures the total amount of redundant joint motion that occurred during the commanded disturbance interval. Its unit is radians and the reported values are converted to degrees. For example, motion of $+5^\circ$ along a redundant direction followed by -5° back along the same direction contributes approximately 10° to E_N , even though the final redundant configuration can be close to its starting value. In this sense E_N is a joint-space path-length quantity, not a torque or energy measure.

The second quantity retains the direction of the motion. Integrating the projected velocity without taking its norm gives the accumulated projected joint displacement

$$\Delta q_{\text{null}} = \int_{5s}^{9s} \dot{q}_{\text{null}}(t) dt = \int_{5s}^{9s} N_q(q(t)) \dot{q}(t) dt. \quad (4.13)$$

The vector $\Delta q_{\text{null}} \in \mathbb{R}^7$ has units of radians. It describes the net accumulated projected joint motion during the disturbance interval, so motion in opposite directions can cancel. It is not an integrated torque: the integrated quantity is joint velocity.

A common direction is introduced to express this seven-dimensional net displacement by one signed scalar. Let $\Delta q_{\text{null},0}$ denote the arithmetic mean of the three net projected displacement vectors recorded in the condition without null-space torque. The corresponding unit direction is

$$v_{\text{ref}} = \frac{\Delta q_{\text{null},0}}{\|\Delta q_{\text{null},0}\|_2}. \quad (4.14)$$

Thus v_{ref} represents the redundant joint-space direction in which the commanded virtual disturbance moves the robot in the reference condition without a secondary null-space torque. The same direction is used for all tested null-space settings. Their signed net displacement along this common direction is

$$\Delta \eta_{\text{dist}} = v_{\text{ref}}^\top \Delta q_{\text{null}}. \quad (4.15)$$

A positive $\Delta \eta_{\text{dist}}$ denotes net motion in the same direction as the reference disturbance

response, a negative value denotes net motion in the opposite direction, and a value near zero indicates little remaining net displacement along that direction. The quantity is a signed projection of the accumulated projected joint motion rather than an exact nonlinear redundant-coordinate displacement, because $N_q(q)$ changes with the robot configuration while v_{ref} is held fixed. At full row rank the Panda has a one-dimensional instantaneous null space, so this common direction provides a meaningful signed comparison of the four tested settings.

The two redundant-motion metrics therefore answer different questions: E_N measures how much redundant joint motion occurred in total, whereas $\Delta\eta_{\text{dist}}$ measures how much net displacement remained along the common reference disturbance direction. A trajectory can consequently have a comparatively large E_N while ending close to its starting redundant configuration and therefore having a small $\Delta\eta_{\text{dist}}$.

Singular-value conditioning was evaluated from the change in σ_{min} across the same disturbance interval, and Cartesian position retention from the peak position-error norm. The corresponding results are presented in Chapter 5.

Chapter 5

Results and Discussion

The surface-contact experiments evaluate how the controller parameters influence the end-effector rotation generated during contact set-up. The primary response quantity is the signed set-up response γ_{t_i} about the investigated surface tangent t_i , defined in Section 4.5. It represents the end-of-set-up orientation change relative to the orientation reference held through set-up. Its sign is chosen so that a response reducing a commanded offset carries the same sign as that offset, and $|\gamma_{t_i}|$ is then the size of the correction.

The commanded tool orientation offset θ_{t_i} defines the initial experimental condition, whereas γ_{t_i} is the measured response. The main response figures therefore report γ_{t_i} , while the corresponding numerical values are provided in Appendix D. The pose-based alignment estimate and the TCP-height flatness classification are retained only as supporting checks in the same appendix.

Each contact parameter setting was repeated three times. The comparative response values in Cases A–D are the arithmetic means over those repetitions, and the corresponding sample standard deviations are given in Appendix D. Repetition scatter is summarised once for each main case below. Within these cases, individual values are discussed only where their spread is relevant to the interpretation. The Case-D comparison additionally carries error bars. The preceding orientation phase terminated with a finite tracking tolerance, so the initial angular magnitude differed slightly from the commanded value. A commanded $+10^\circ$ offset reached approximately 9.3° about t_1 and 10.0° about t_2 . The commanded offset identifies each experimental condition.

In the response figures that follow, γ_{t_i} carries the same sign as the commanded offset

when the response reduces that offset, and its magnitude is the size of the correction. A response of the opposite sign to the commanded offset moved the end effector further from the flat orientation. Both signs therefore have to be read against the commanded offset of the condition, which Section 4.5.1 defines.

5.1 Main Surface-Contact Results

The cases are reported in the order in which they constrain the choice of compliance-centre position. The TCP-centred reference is reported in Case A, followed by the stiffness sensitivities in Cases B and C. In Case D the tangential compliance-centre displacement was tested over both surface tangents and both signs of the commanded orientation offset. The orientation-magnitude, definition-frame, tool-axis, and intermediate-direction comparisons are reported as supporting checks in Appendix D.

5.1.1 Baseline Contact Response with the CoC at the TCP: Case A

In Case A the CoC was placed at the TCP, so $r_c = 0$. This condition removes the additional translation–rotation coupling introduced by the virtual point shift. The ordinary rotational impedance and physical contact geometry remain. The measured response therefore provides the baseline for the stiffness and compliance-centre-position comparisons. Repetition scatter remained at or below 0.19° across all Case-A settings in Table D.1.

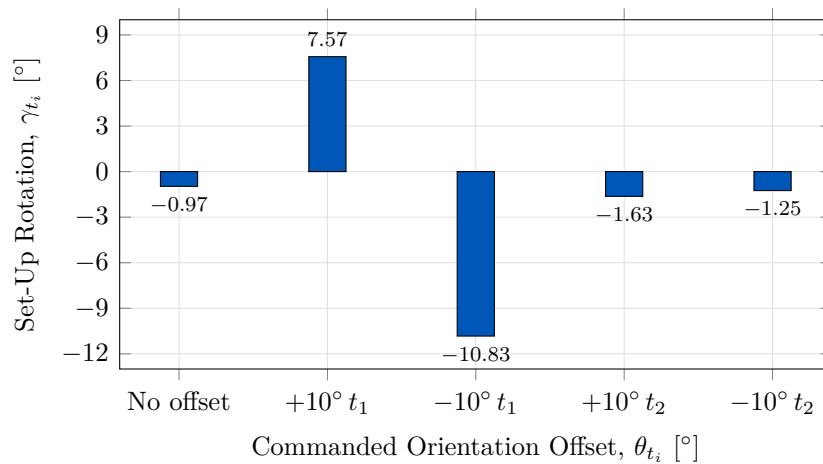


Figure 5.1: Measured set-up rotation with the CoC at the TCP.

With the CoC at the TCP, the measured response differs strongly between the two surface tangents. For the positive 10° conditions, Figure 5.1 gives $+7.57^\circ$ about t_1 and -1.63° about t_2 . Reversing the commanded orientation changes the response to -10.83° and -1.25° , respectively. The figure shows all five baseline conditions.

The condition without a commanded orientation offset gave -0.97° about t_1 . Contact produced end-effector rotation from the flat target orientation as well. The commanded conditions are read against that non-zero response rather than against zero. The TCP-centred condition provides the reference against which the stiffness and compliance-centre-position variations are compared.

5.1.2 Effect of Cartesian Stiffness: Cases B and C

Rotational Stiffness: Case B

In Case B the rotational stiffness about the investigated tangent was varied while the other controller settings remained unchanged. The 5 N m/rad condition is the corresponding TCP-centred reference from Case A. Repetition scatter remained at or below 0.31° across all Case-B settings in Table D.2.

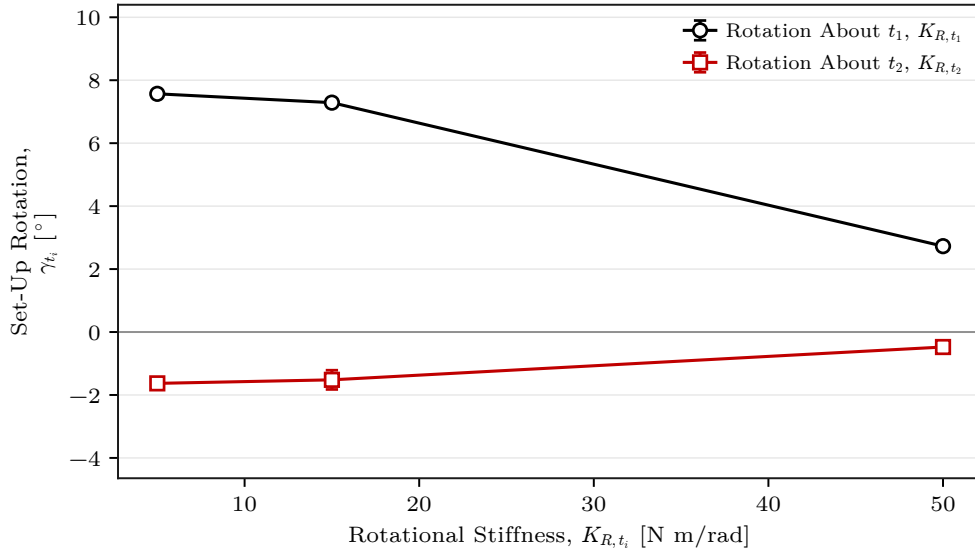


Figure 5.2: Measured set-up rotation by rotational stiffness.

Increasing K_{R,t_i} reduced the measured rotation over the tested range. About t_1 , Figure 5.2 shows a decrease from $+7.57^\circ$ at 5 N m/rad to $+2.73^\circ$ at 50 N m/rad . About t_2 , the re-

sponse remained small and changed from -1.63° to -0.48° . Rotational stiffness therefore changed the response magnitude without removing the directional difference observed in Case A.

Cross-Axis Translational Stiffness: Case C

In Case C the translational stiffness perpendicular to the investigated rotation axis was varied. The rotational stiffness about that axis was held at 5 N m/rad . The 2000 N/m condition is the corresponding TCP-centred reference from Case A. Repetition scatter remained at or below 0.17° across all Case-C settings in Table D.3.

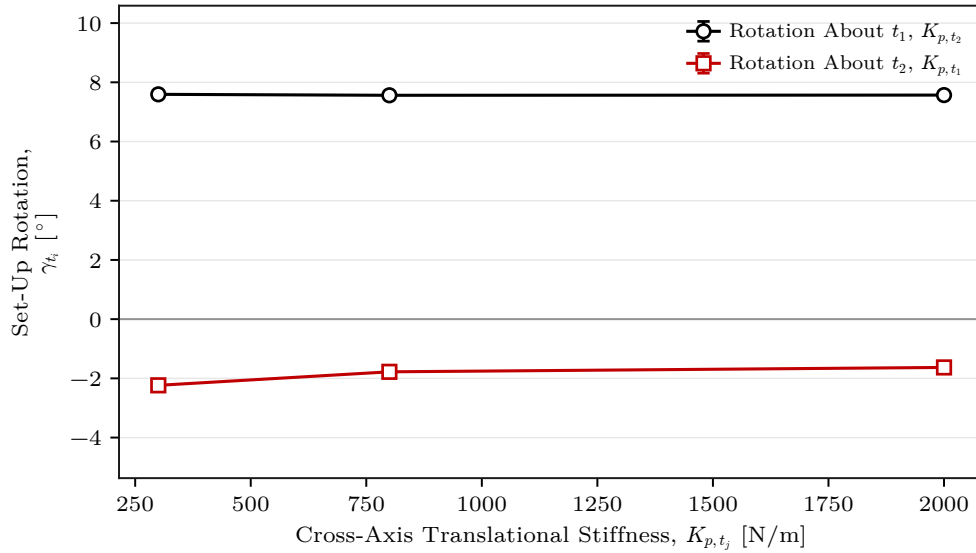


Figure 5.3: Measured set-up rotation by cross-axis translational stiffness.

The influence of perpendicular translational stiffness was considerably smaller than that of rotational stiffness. About t_1 , the measured rotation remained within 0.03° across the tested range. About t_2 , Figure 5.3 shows a change of 0.61° , from -2.24° to -1.63° . Varying this stiffness alone therefore did not account for the directional difference in the baseline response.

5.1.3 Effect of Tangential CoC Position: Case D

In Case D the tangential position of the CoC was varied while the Cartesian gains remained unchanged. For each rotation axis, the centre was moved along the surface tangent

perpendicular to that axis. The TCP-centred condition is included as the zero-position reference. Repetition scatter was small at most Case-D settings, and Table D.4 gives every sample standard deviation. Four conditions carry a larger spread, and they are named where the error bars appear below rather than summarised by a single bound.

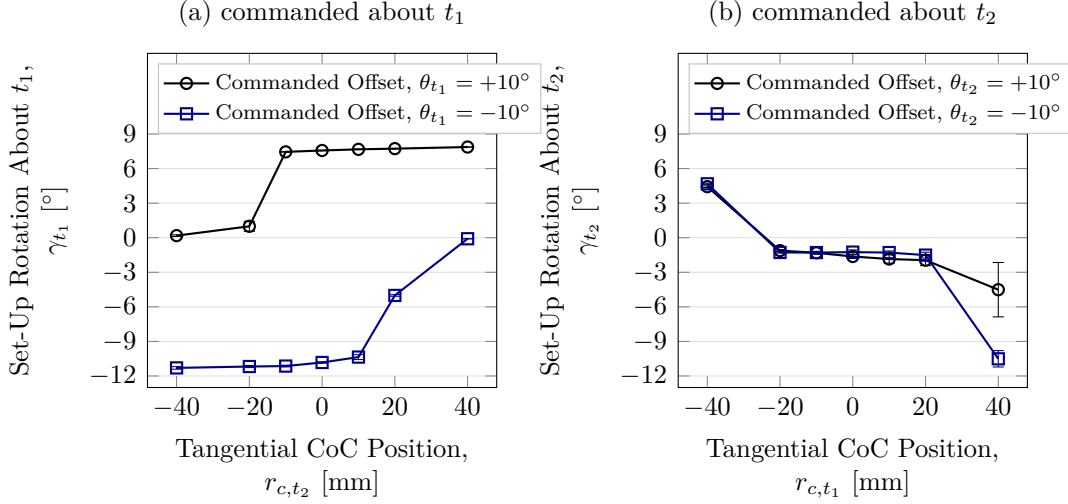


Figure 5.4: Set-up rotation by tangential CoC position (mean $\pm$ SD).

Tangential CoC position produced the largest response variation among the tested controller parameters. As shown in Figure 5.4, reversing the initial orientation condition changes which side of the TCP produces the larger response. About t_1 , the positive condition gives $+7.87^\circ$ at $+40$ mm and $+0.18^\circ$ at -40 mm. The reversed condition exchanges the larger response between the two sides.

Changing the rotation axis also changes the required tangential direction. For a positive condition about t_2 , the response changes from -1.63° at the TCP to $+4.43^\circ$ at the selected $r_{c,t_1} = -40$ mm position. The selected displacement therefore lies along $+t_2$ for rotation about t_1 and along $-t_1$ for rotation about t_2 . Within the investigated configuration, a non-zero tangential CoC therefore cannot be selected independently of the direction and sign of the initial angular condition.

The error bars in Figure 5.4 identify the conditions with the larger repetition spread. The positive t_1 condition at -20 mm had a standard deviation of 0.44° . About t_2 , the positive condition had standard deviations of 0.47° at $+20$ mm and 2.36° at $+40$ mm. The reversed condition at $+40$ mm had a standard deviation of 0.71° .

The positive t_1 condition was examined in the time domain to relate the rotation response to the commanded wrench. Its two outer positions produce a large difference in γ_{t_1} , while

the TCP-centred condition provides the zero-position reference. Each trace is one recorded repetition of its centre position.

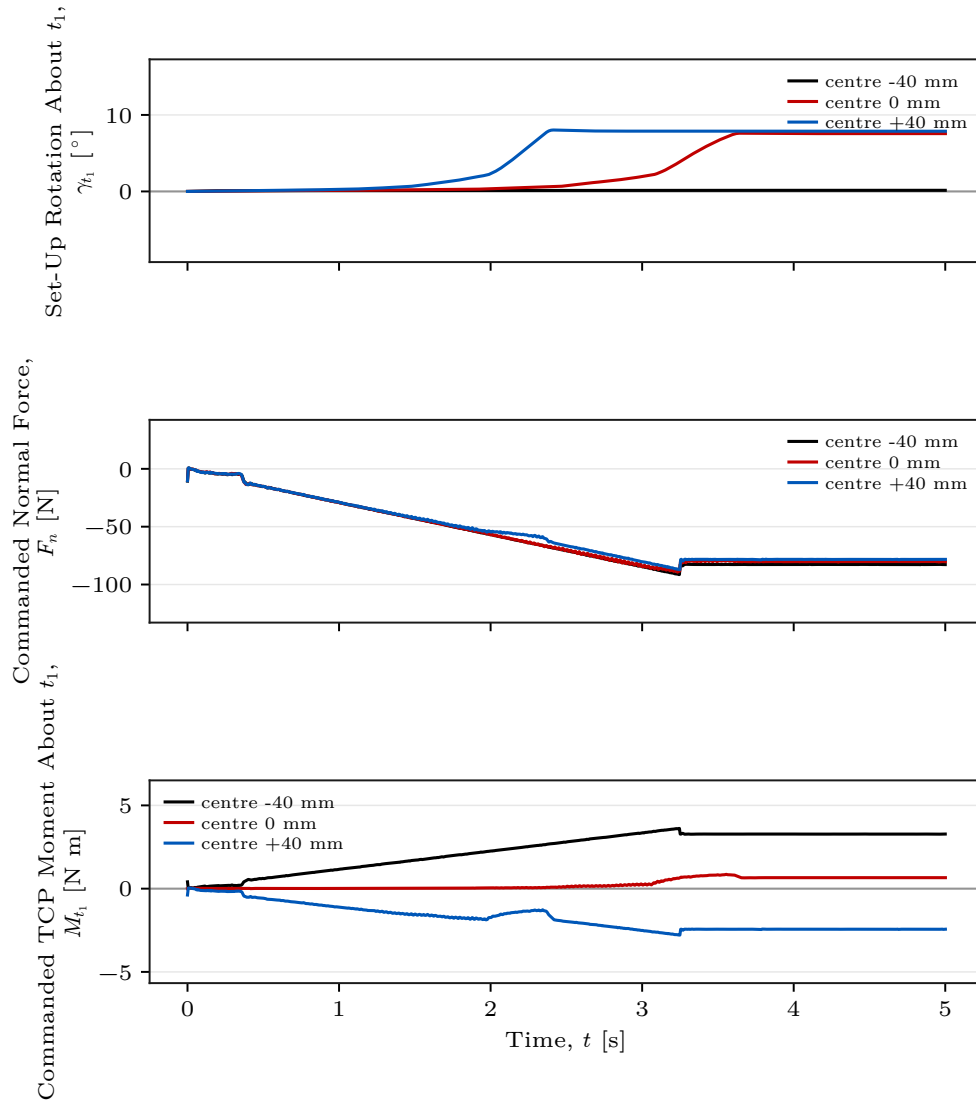


Figure 5.5: Measured set-up rotation, commanded normal force, and commanded TCP moment for three tangential CoC positions.

Figure 5.5 places the set-up rotation γ_{t_1} , the commanded normal force F_n , and the commanded TCP moment M_{t_1} on a common time axis for three centre positions. Every condition shown carries a commanded orientation offset of $+10^\circ$ about t_1 . At -40 , 0 , and $+40$ mm, the commanded normal force settles in a similar range of approximately -77 to -82 N, while the commanded TCP moment about t_1 settles at approximately $+3.28$, $+0.66$, and -2.44 N m. The outer positions therefore produce moments of similar magnitude and opposite sign while their set-up rotations differ by 7.73° . The difference in set-up rotation is therefore associated primarily with the moment generated by the dis-

placed impedance reference rather than with a substantial change in the normal pressing force.

The commanded moment and the reported rotation carry opposite signs here. γ_{t_1} is the rotation from the measured end-effector orientation back to the reference held through set-up, as Section 4.5.1 defines it. Its sign is therefore opposite to that of the physical rotation. The alignment-directed response at +40 mm appears as a positive γ_{t_1} against a negative commanded M_{t_1} .

A tangential displacement of the CoC moves the impedance reference point away from the TCP, so the translational press is coupled to the rotational part of the impedance. The complete commanded moment is $m = m_R + r_c \times f$, with $r_c = p_c - p_{\text{TCP}}$. A non-zero tangential displacement therefore adds a moment contribution from the commanded press. This explains why the commanded moment changes strongly with centre position while F_n remains within a similar range. Reversing the tangential displacement reverses the contribution $r_c \times f_n$ generated by the normal press.

The measured sign reversal follows the relation derived in Section 2.7.2. A fixed tangential $r_c \neq 0$ retains one preferred moment direction while the press continues. At $r_c = 0$, the point-shift contribution vanishes and the TCP selects no tangential direction. The TCP-centred condition is therefore the direction-independent fixed reference among the tested positions. A displaced centre provides greater condition-specific rotational authority where the required direction is known.

The grinding phase returns to the decoupled impedance branch described in Section 3.2.5. This phase structure is consistent with removing the direction-selective coupling after set-up. It is an implementation property rather than an experimental result, because the reported runs ended at the pre-grinding gate.

Across Cases A–D, tangential CoC position produced the largest measured response variation among the tested controller parameters.

5.2 Null-Space Pose-Hold Results

The remaining experiment is separate from the contact cases and analyses the redundant joint-space direction while the Cartesian pose is held. Matched trials compare the

selectable null-space terms under the same internally commanded point-force-equivalent disturbance. The commanded disturbance reached 20.00 N in all twelve trials, with peak equivalent joint-torque norms between 2.17 and 2.45 N m. The torque-limit scale remained equal to one, so the waveform was applied without clipping.

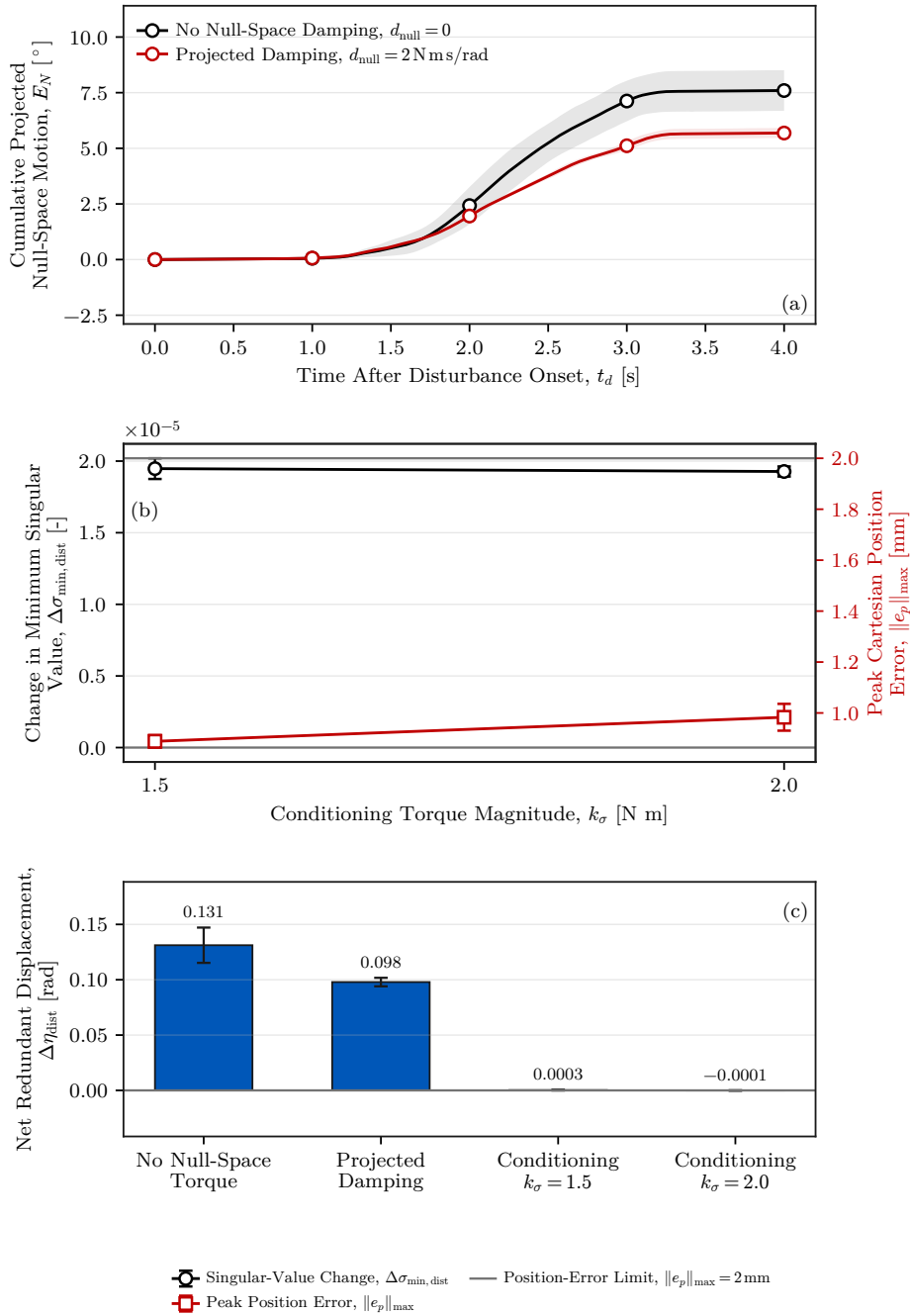


Figure 5.6: Null-space pose-hold response.

Figure 5.6 reports the response during the Cartesian pose-hold disturbance test in three panels. Panel (a) gives the mean cumulative projected null-space motion E_N without

null-space torque and with projected damping, the shaded bands showing $\pm$ one standard deviation across the three repetitions. Panel (b) gives the change in the minimum Jacobian singular value over the disturbance interval, $\Delta\sigma_{\min,\text{dist}}$, together with the peak Cartesian position error, for the two singular-value conditioning magnitudes; its error bars show $\pm$ one standard deviation and the horizontal line marks the 2 mm position-retention limit. Panel (c) gives the net redundant displacement $\Delta\eta_{\text{dist}}$ of all four settings, with the mean printed above each bar and error bars showing $\pm$ one standard deviation. The complete numerical values are given below.

5.2.1 Projected Null-Space Damping

Projected null-space damping reduced E_N from 7.596° without null-space torque to 5.687° at $d_{\text{null}} = 2 \text{ N m s/rad}$, a reduction of approximately 25 %; panel (a) of Figure 5.6 carries both traces. The net redundant displacement decreased in the same proportion, from 0.131 rad to 0.098 rad, while the change in $\sigma_{\min}$ across the disturbance interval became less negative, from -2.13×10^{-3} to -1.26×10^{-3} . Damping acts on the projected joint velocity through

$$\tau_d = -d_{\text{null}} N_\tau \dot{q},$$

so it vanishes as the redundant motion stops and requests no return towards the initial configuration.

5.2.2 Singular-Value Conditioning

Singular-value conditioning behaved differently from both the uncontrolled and the damping-only settings. It was already active during the 5 s preceding the disturbance and remained active throughout it, so it did not act as a term applied after the redundant configuration had been displaced. Both sigma-only settings ended the disturbance interval with a net redundant displacement close to zero: 0.3×10^{-3} rad at $k_\sigma = 1.5 \text{ N m}$ and -0.1×10^{-3} rad at 2.0 N m . As panel (c) of Figure 5.6 shows, both values are more than two orders of magnitude smaller than the 0.131 rad obtained without null-space torque. Over the same interval $\sigma_{\min}$ increased by approximately 1.95×10^{-5} and 1.93×10^{-5} ,

whereas it decreased by 2.13×10^{-3} without null-space torque. Under the tested disturbance direction, the conditioning term therefore strongly suppressed the resulting redundant displacement.

The two sigma magnitudes differed mainly in motion activity. At $k_\sigma = 2.0 \text{ N m}$, the cumulative projected null-space motion was 1.619° , compared with 0.283° at 1.5 N m , a factor of about six, while both net displacements remained near zero. The larger torque magnitude was associated with greater switching activity near the locally preferred region. This is consistent with the control law, because the conditioning term commands a fixed torque magnitude along the selected direction and the selector s_σ becomes zero only inside the comparison deadband. The larger cumulative motion therefore indicates greater switching activity around the same region rather than poorer disturbance rejection. The mean peak Cartesian position error was 0.889 mm at 1.5 N m and 0.983 mm at 2.0 N m . The largest position error measured in the complete pose-hold study was 1.304 mm , below the predefined 2 mm acceptance limit. Of the two tested magnitudes, 1.5 N m achieved comparable suppression of net disturbance-induced displacement with substantially less redundant motion.

5.2.3 Interpretation and Experimental Bounds

The pose-hold trials compare the four selectable null-space modes under the repeatable commanded disturbance. The two terms act on different quantities. Projected damping dissipates redundant joint motion that is already present, whereas the singular-value term drives the redundant configuration towards the sampled neighbour with the larger $\sigma_{\min}$ whether or not such motion is present. The conditioning law is not dissipative: a disturbance acting towards larger $\sigma_{\min}$ would move the configuration in the same direction as τ_σ rather than against it, so the rejection observed here is specific to the tested disturbance direction. Both terms act through the null-space projector, so neither contributes a first-order Cartesian wrench.

Several bounds apply to these results. Because the conditioning torque was active before the disturbance, the sigma-only trials entered it from a redundant configuration displaced by approximately 0.013 rad from the one at which the trials without conditioning began. The four settings therefore compare the closed-loop behaviour of the

complete modes. They do not isolate an instantaneous opposing torque at identical joint configurations. The surface-contact experiments use both contributions together in the combined secondary-torque mode and therefore do not separate their individual effects during contact. The geometric Jacobian combines translational and rotational rows without a common length scale, so the reported $\sigma_{\min}$ is used only as a relative indicator under the fixed Jacobian convention.

Chapter 6

Conclusion and Future Work

6.1 Conclusion

A real-time Cartesian impedance controller was implemented for contact-induced alignment of a rectangular tool with a calibrated plane. The controller assigns directional gains in the surface frame and can shift the virtual centre of compliance from the TCP. The shift couples the commanded press to the rotational impedance through $m = m_R + r_c \times f$.

The TCP-centred conditions established a direction-dependent contact response. Substantial alignment-directed end-effector rotation was measured about t_1 , while the response about t_2 was smaller and had the opposite sign for the positive command. Raising the rotational stiffness reduced the response over the tested range. Changing the cross-axis translational stiffness produced a smaller effect.

Tangential compliance-centre position produced the largest measured variation. The displacement that increased alignment depended on the sign of the initial orientation and on the commanded rotation axis. Reversing the displacement reversed the moment contribution from the normal press. Within the investigated contact configuration and tested positions, a non-zero tangential centre could therefore not be selected independently of the direction and sign of the initial misalignment.

The TCP-centred impedance is the direction-independent fixed centre of compliance among the tested positions, because it selects no tangential direction. A displaced centre provides greater condition-specific rotational authority when the required direction is known. These properties are distinct: the fixed reference need not produce the largest

alignment-directed rotation in every condition.

The null-space pose-hold study separated projected joint damping from singular-value conditioning. Damping reduced redundant joint motion while the disturbance acted. Under the tested disturbance direction, conditioning strongly suppressed the net redundant displacement and kept the Cartesian position error below the specified 2 mm limit. The lower tested conditioning magnitude produced the same suppression with less cumulative redundant motion.

6.2 Limitations

6.2.1 Experimental Range

The contact study used one Panda configuration, one tool and mounting arrangement, one calibrated plane, and one press trajectory. The conclusions are therefore limited to this geometry and the tested stiffness, orientation-offset, and compliance-centre ranges. Sustained grinding and a centre scheduled during contact were not evaluated.

6.2.2 Measurement and Tool-Mount Uncertainty

The primary response was calculated from the measured end-effector rotation. The tool mount exhibited approximately $\pm 2^\circ$ of unloaded play about y_{EE} , and its instantaneous motion was not tracked during contact. The results therefore describe end-effector rotation and do not provide an independent physical tool-face angle.

6.2.3 Null-Space Configuration and Real-Time Operation

The null-space terms were separated only in Cartesian pose hold under one internally commanded disturbance direction. The conditioning torque was active before the disturbance, so the experiment compares complete closed-loop modes rather than isolated torque contributions at identical configurations. The surface-contact study held the combined mode fixed. Worst-case execution time and scheduling jitter were not measured separately from operation through the nominal 1 kHz interface.

6.3 Future Work

The first priority is an independent measurement of the physical tool-face angle during contact. This would separate end-effector rotation from motion in the tool mount and provide a direct alignment reference.

The second priority is a direction-selected compliance-centre strategy. Such a controller would begin at the TCP, estimate the required alignment direction, apply a bounded tangential displacement during set-up, and return to the TCP before sustained contact. Negative orientation offsets and additional tangent-plane directions should be included in its evaluation.

The third priority is a combined-mode null-space study under physical disturbance. Varying projected damping and conditioning together would show whether the pose-hold findings remain valid during contact. A continuous or hysteretic conditioning law should also be compared with the present two-point selector to reduce its switching activity.

Bibliography

- [1] Alin Albu-Schäffer, Christian Ott, Udo Frese, and Gerd Hirzinger. Cartesian impedance control of redundant robots: Recent results with the DLR-light-weight-arms. In *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*, volume 3, pages 3704–3709, 2003. doi: 10.1109/ROBOT.2003.1242165.
- [2] Benjamin Alt, Florian Stöckl, Silvan Müller, Christopher Braun, Julian Raible, Saad Alhasan, Oliver Rettig, Lukas Ringle, Darko Katic, Rainer Jäkel, Michael Beetz, Marcus Strand, and Marco F. Huber. RoboGrind: Intuitive and interactive surface treatment with industrial robots. In *2024 IEEE International Conference on Robotics and Automation (ICRA)*, pages 2140–2146, 2024. doi: 10.1109/ICRA57147.2024.10611143.
- [3] John J. Craig. *Introduction to Robotics: Mechanics and Control*. Pearson Prentice Hall, Upper Saddle River, NJ, 3 edition, 2005.
- [4] Jacques Denavit and Richard S. Hartenberg. A kinematic notation for lower-pair mechanisms based on matrices. *Journal of Applied Mechanics*, 22(2):215–221, 1955.
- [5] Ernest D. Fasse and Jan F. Broenink. A spatial impedance controller for robotic manipulation. *IEEE Transactions on Robotics and Automation*, 13(4):546–556, 1997. doi: 10.1109/70.611320.
- [6] Franka Emika GmbH. libfranka: Communication Test Example. https://github.com/frankarobotics/libfranka/blob/0.7.1/examples/communication_test.cpp, 2019. Official libfranka 0.7.1 example, file `examples/communication_test.cpp`.
- [7] Franka Emika GmbH. Franka control interface documentation. <https://frankaemika.github.io/docs/>, 2023. Accessed for robot interface, libfranka model access, and torque control documentation.

- [8] Franka Emika GmbH. libfranka: Cartesian Impedance Control Example. https://github.com/frankaemika/libfranka/blob/master/examples/cartesian_impedance_control.cpp, 2023. Official libfranka example, file `examples/cartesian_impedance_control.cpp`.
- [9] Christian Friedrich, Patrick Frank, Marco Santin, and Matthias Haag. Highly dynamic physical interaction for robotics: Design and control of an active remote center of compliance. In *2025 IEEE International Conference on Robotics and Automation (ICRA)*, 2025. doi: 10.1109/ICRA55743.2025.11128451.
- [10] Richard Hartisch and Kevin Haninger. Flexure-based environmental compliance for high-speed robotic contact tasks. In *2022 IEEE/ASME International Conference on Advanced Intelligent Mechatronics (AIM)*, pages 1608–1613, 2022. doi: 10.1109/AIM52237.2022.9863334.
- [11] Neville Hogan. Impedance control: An approach to manipulation—parts i–iii. *Journal of Dynamic Systems, Measurement, and Control*, 107(1):1–24, 1985.
- [12] International Federation of Robotics. World robotics 2023 report. Technical report, IFR Statistical Department, Frankfurt am Main, 2023.
- [13] Oussama Khatib. A unified approach for motion and force control of robot manipulators. *IEEE Journal on Robotics and Automation*, 3(1):43–53, 1987.
- [14] Kevin M. Lynch and Frank C. Park. *Modern Robotics: Mechanics, Planning, and Control*. Cambridge University Press, 2017. URL <https://hades.mech.northwestern.edu/images/7/7f/MR.pdf>. Free preprint.
- [15] Matthias Mayr and Julian M. Salt-Ducaju. A C++ implementation of a cartesian impedance controller for robotic manipulators. *Journal of Open Source Software*, 9(93):5194, 2024. doi: 10.21105/joss.05194.
- [16] Kang Min, Fenglei Ni, Zhaoyang Chen, and Hong Liu. A force control method integrating human skills for complex surface finishing. *Machines*, 12(11):756, 2024. doi: 10.3390/machines12110756.
- [17] Richard M. Murray, Zexiang Li, and S. Shankar Sastry. *A Mathematical Introduction to Robotic Manipulation*. CRC Press, 1994.

- [18] James L. Nevins and Daniel E. Whitney. Computer-controlled assembly. *Scientific American*, 1989. Background source on robotic assembly and compliance.
- [19] Christian Ott, Alin Albu-Schäffer, Andreas Kugi, and Gerd Hirzinger. On the passivity-based impedance control of flexible joint robots. *IEEE Transactions on Robotics*, 24(2):416–429, 2008.
- [20] Mathew Jose Pollayil, Franco Angelini, Guiyang Xin, Michael Mistry, Sethu Vijayakumar, Antonio Bicchi, and Manolo Garabini. Choosing stiffness and damping for optimal impedance planning. *IEEE Transactions on Robotics*, 39(2):1281–1300, 2023. doi: 10.1109/TRO.2022.3216078.
- [21] Bruno Siciliano, Lorenzo Sciavicco, Luigi Villani, and Giuseppe Oriolo. *Robotics: Modelling, Planning and Control*. Springer, London, 2009.
- [22] Stefano Stramigioli, Claudio Melchiorri, and Stefano Andreotti. Intrinsically passive grasping and manipulation. *IEEE Transactions on Robotics and Automation*, 2000. URL <http://robby.caltech.edu/~jwb/courses/CDS270/papers/passive.grasp.pdf>. Open-access preprint.
- [23] Markku Suomalainen, Sylvain Calinon, Emmanuel Pignat, and Ville Kyrki. Improving dual-arm assembly by master-slave compliance. In *2019 International Conference on Robotics and Automation (ICRA)*, pages 8676–8682, 2019. doi: 10.1109/ICRA.2019.8793977.
- [24] Valeria Villani, Francesco Pini, Francesco Leali, and Cristian Secchi. Survey on human–robot collaboration in industrial settings: Safety, intuitive interfaces and applications. *Mechatronics*, 55:248–266, 2018. doi: 10.1016/j.mechatronics.2018.02.009.
- [25] Sen Wang, Guodong Chen, Hui Xu, and Zheng Wang. A robotic peg-in-hole assembly strategy based on variable compliance center. *IEEE Access*, 2019. doi: 10.1109/ACCESS.2019.2954459.
- [26] Paul C. Watson. Remote center compliance system. U.S. Patent 4,098,001, 1978. Assigned to The Charles Stark Draper Laboratory.

- [27] Chong Wu, Kai Guo, and Jie Sun. Dual PID adaptive variable impedance constant force control for grinding robot. *Applied Sciences*, 13(21):11635, 2023. doi: 10.3390/app132111635.

Appendix A

Controller Source Listings

The following listings show the implementation of the compliance-centre transformation and the final torque assembly. They use fixed-size Eigen aliases such as `Vec3`, `Vec7`, `Mat3`, `Mat6x6` and `Mat6x7`, and `R_base_surface` denotes the surface-frame orientation used in the chapters.

A.1 Centre of Compliance and the Offset Adjoint

The lever $r_c = p_c - p_{\text{TCP}}$ points from the TCP to the selected centre of compliance, and the same congruence is applied to stiffness and damping.

Listing A.1: TCP-to-compliance-centre adjoint and congruence transform.

```
1 | Mat3 skewMatrix(const Vec3& v) {
2 |     Mat3 s;
3 |     s << 0.0, -v(2), v(1),
4 |         v(2), 0.0, -v(0),
5 |         -v(1), v(0), 0.0;
6 |     return s;
7 | }
8 |
9 | Mat6x6 complianceShiftMatrix(const Vec3& r_c) {
10 |     Mat6x6 adjoint = Mat6x6::Zero();
11 |     adjoint.block<3, 3>(0, 0) = Mat3::Identity();
12 |     adjoint.block<3, 3>(0, 3) = -skewMatrix(r_c);
13 |     adjoint.block<3, 3>(3, 3) = Mat3::Identity();
14 |     return adjoint;
15 | }
16 |
17 | Mat6x6 shiftGainToTcp(
18 |     const Mat6x6& gain_at_center, const Vec3& r_c) {
19 |     const Mat6x6 adjoint = complianceShiftMatrix(r_c);
20 |     return adjoint.transpose() * gain_at_center * adjoint;
21 | }
```

Both configured definitions store r_c itself, so the two branches of the wrench routine differ only in the frame it is expressed in.

Listing A.2: Selection of the decoupled or coupled spring wrench (abridged).

```

1 | const bool coupled =
2 |     params.use_virtual_compliance_center &&
3 |     (phase == ControlPhase::kSetup ||
4 |     (phase == ControlPhase::kPoseHold &&
5 |     params.use_setup_impedance_hold));
6 | if (!coupled) {
7 |     // Two independent springs. dv.tail is -omega, so the damping
8 |     // signs match.
9 |     Vec6 wrench;
10 |    wrench.head<3>() = Kp * dx.head<3>() + Dp * dv.head<3>();
11 |    wrench.tail<3>() = KR * dx.tail<3>() + DR * dv.tail<3>();
12 |    return wrench;
13 | }
14 |
15 | Vec3 r_c = Vec3::Zero();
16 | if (params.compliance_center_in_tool_frame) {
17 |     r_c = R_EE * params.compliance_center_offset_ee;
18 | } else {
19 |     r_c = R_base_surface * params.compliance_lever_surface;
20 | }
21 | const Mat6x6 K_tcp =
22 |     shiftGainToTcp(blockDiagonal(Kp, KR), r_c);
23 | const Mat6x6 D_tcp =
24 |     shiftGainToTcp(blockDiagonal(Dp, DR), r_c);
25 | return K_tcp * dx + D_tcp * dv;
    
```

A.2 Core Control Callback

The two listings below show the state and model quantities acquired at the start of each torque callback and how they enter the torque command.

Listing A.3: Robot-state and model quantities acquired by the torque callback.

```

1 | Map<const Vec7> q_current(state.q.data());
2 | Map<const Vec7> dq(state.dq.data());
3 |
4 | const auto jacobian_array =
5 |     model.zeroJacobian(Frame::kEndEffector, state);
6 | Map<const Mat6x7> J(jacobian_array.data());
7 | const Vec6 xdot = J * dq;
8 |
9 | Map<const Mat4x4> T_EE(state.O_T_EE.data());
10 | const Vec3 p_EE = T_EE.block<3, 1>(0, 3);
11 | const Mat3 R_EE = T_EE.block<3, 3>(0, 0);
    
```

Listing A.4: Core impedance, coupled set-up, and torque composition.

```

1 | const Vec3 e_p = desired.p_d - p_EE;
2 | const Vec3 e_R = applyRotationalAxisMask(
3 |     params, orientationError(R_EE, R_d_used), R_base_surface);
4 |
5 | Vec6 dx;
6 | dx.head<3>() = e_p;
7 | dx.tail<3>() = e_R;
8 | Vec6 dv;
9 | dv.head<3>() = desired.pdot_d - pdot;
10 | dv.tail<3>() = -omega;
11 |
12 | const Vec6 wrench =
13 |     computeCartesianImpedanceWrench(
14 |         params, phase, Kp_used, Dp_used,
15 |         KR_used, DR_used, R_base_surface,
16 |         dx, dv, R_EE);
17 |
18 | const Vec7 tau_nullspace = computeNullspaceTorque(
19 |     params, model, state, J, dq);
    
```

```

20 |
21 | AutomaticDisturbance disturbance;
22 | if (phase == ControlPhase::kPoseHold &&
23 |     params.disturbance_auto_enabled) {
24 |     disturbance = computeAutomaticDisturbance(
25 |         params, model, state, disturbance_force_direction_base,
26 |         time - phase_start_time);
27 | }
28 |
29 | Array7 coriolis_array = model.coriolis(state);
30 | Map<const Vec7> coriolis(coriolis_array.data());
31 | const Vec7 tau_task = J.transpose() * wrench;
32 | const Vec7 tau_cmd =
33 |     tau_task + tau_nullspace + disturbance.tau + coriolis;

```

The disturbance term is non-zero only in the Cartesian pose-hold study of Section 4.6 and is zero in every surface-contact run.

Appendix B

Recorded Experimental Data

The controller stores experimental signals in a preallocated ring buffer. This avoids file access and dynamic buffer growth inside the 1 kHz control callback. After the controller stops, the retained rows are written in chronological order to a CSV file.

B.1 Control Phases

The integer `phase` identifies the active part of the experiment:

phase	0	1	2	3	4	5
state	<code>approach_orient</code>	<code>approach_descend</code>	<code>set_up</code>	<code>grind</code>	<code>hold</code>	<code>manual_guide.</code>

B.2 Recorded Signals

Table B.1 lists the recorded signal groups relevant to the experiments and subsequent evaluation. A suffix `_x, _y, _z` denotes base-frame components, and `_1 . . _7` denotes joint components.

Table B.1: Recorded signal groups.

Column group	Meaning
<code>time, phase</code>	Run time and active control state.
<code>p_EE, p_d</code>	Measured and desired TCP positions [m].

Table B.1 continued

Column group	Meaning
tool_contact, edge_target	Selected contact point and its reference [m].
tool_contact_offset_ee	Offset of the selected contact point from the TCP, in EE coordinates [m].
first_contact_tcp, first_contact	TCP position and projected surface reference captured at the clearance transition [m].
e_p, e_R, pdot, pdot_d, omega	Cartesian errors and velocities; during set-up e_R is referred to the orientation held from the clearance transition.
angular_deviation_deg	Pose-based alignment error $\theta_{\text{align}} = \cos^{-1}(-n_{\text{Tool},0}^{\top} n_s)$, used in the consistency check of Appendix D.7 [°].
angular_deviation_t1_deg, angular_deviation_t2_deg, angular_deviation_ normal_deg	The same angular deviation resolved on t_1 , t_2 and n_s [°]. Not evaluation quantities.
p_CoC, r_eff	CoC $p_{\text{TCP}} + r_c$ and the offset from it to the selected tool point, $p_{\text{Tool}} - p_c$, in the base frame [m]. Written for every run, including $r_c = 0$.
t_align	Time from the start of set-up at which the deviation first settled inside the configured tolerance [s], negative until then. Observed only; it does not end the phase.
f, m	Commanded Cartesian force [N] and moment [N m].
external_force, external_moment	Model-estimated external wrench from libfranka, expressed in the base orientation and referenced to stiffness frame K .
contact_force_bias, contact_moment_bias	External-wrench values captured at the clearance transition.
force_after_contact, moment_after_contact	The model-estimated wrench with those reference values already subtracted.

Table B.1 continued

Column group	Meaning
push	Set-up penetration reference or grind preload retained after set-up [m].
nullspace_mode, sigma_current	Selected null-space law and current smallest singular value $\sigma_{\min}$.
sigma_plus, sigma_minus, sigma_difference, sigma_direction	$\sigma_{\min}$ at the two sampled configurations $q \pm \alpha_{\text{probe}} v_7$, their difference and the selected sign. Recorded in every mode.
nullspace_dq_1..7	Projected joint velocity $\dot{q}_{\text{null}}$ [rad/s].
tau_sigma_1..7, tau_sigma_norm, tau_nullspace_norm	Singular-value torque τ_{σ} and total null-space torque τ_{null} [N m].
nullspace_damping	Active projected damping gain [N m s/rad].
disturbance_scale, disturbance_force_ base_x..z, disturbance_ point_base_x..z	Commanded hold-disturbance waveform, point force [N] and the base-frame position of the link point it acts at [m].
tau_disturbance_1..7	Equivalent joint torque $J_p^{\top} f_d$ in the internally commanded hold study [N m].
tau_cmd_1..7	Final commanded joint torque [N m].

Appendix C

Controller Parameters

This appendix lists the effective controller parameters used in the reported experiments. It is a reproducibility record for the implementation described in Chapters 3 and 4.

C.1 Surface and Tool Geometry

The plane satisfies

$$n_s^\top(p - p_s) = 0, \quad n_s = R_y(\beta_s)R_x(\alpha_s)e_{z_0},$$

and the surface-frame column order is $[t_1, t_2, n_s]$. Table C.1 collects the corresponding surface and tool parameters.

Table C.1: Surface and tool parameters.

Quantity	Configuration key	Value
Surface point	<code>surface_point</code>	$[0.515267, -0.107172, 0.003108]$ m
Surface tilts	α_s, β_s	$-1.585^\circ, +0.988^\circ$
First-tangent hint	<code>surface_tangent1_hint_base</code>	$[1, 0, 0]$
Tool axis in EE	<code>tool_axis_ee</code>	$[0.026387057, -0.006678514, 0.999629492]$
Signed alignment target	<code>tool_axis_target_sign</code>	$-n_s$
Spin about the tool axis, from t_1	<code>tool_target_offset_normal_deg</code>	90.0°
Tool-face centre in EE	<code>tool_contact_face_center_ee</code>	$[0, 0, 0.020]$ m
Tool face size	width $\times$ length	0.040×0.120 m
Tool-point selection tolerance, ε_{sel}	<code>tool_contact_feature_tie_tolerance</code>	0.1 mm

C.1.1 Calibration Relations

For the seated surface pose, the nominal end-effector axis and the tool-face centre offset supplied the stored plane quantities,

$$n_s = R_{\text{EE}}[0, 0, 1]^\top, \quad p_s = p_{\text{EE}} + R_{\text{EE}}r_{\text{face,EE}}. \quad (\text{C.1})$$

The first tangent was the normalised projection of the base-frame x direction onto the plane, and $t_2 = n_s \times t_1$ completed the surface frame. For the tool-normal calibration, the fitted direction was

$$n_{\text{Tool,EE}} = \arg \min_{\|n\|=1} \sum_{i < j} \|(R_{\text{cal},i} - R_{\text{cal},j})n\|^2. \quad (\text{C.2})$$

Three seated yaw captures entered the fit. The sign of the fitted eigenvector was selected towards nominal $+Z_{\text{EE}}$.

C.1.2 Initial Joint Configuration

The initial joint configuration used in the reported experiments is

$$q_{\text{init}} = [0.014777, 0.210911, -0.009320, -2.284835, \\ -0.002725, 2.493410, 0.780694]^\top \text{ rad.}$$

C.2 Phase Sequence and Impedance

Table C.2: Phase and mode gains.

Phase / quantity	Active stiffness and frame	Configured damping
Approach translation	[150, 150, 150] N/m, surface $[t_1, t_2, n_s]$	[20, 20, 20] N s/m
Approach rotation	[180, 180, 90] N m/rad, surface $[t_1, t_2, n_s]$	[15, 15, 12] N m s/rad
Set-up translation	[2000, 2000, 350] N/m, surface $[t_1, t_2, n_s]$	[10, 10, 175] N s/m
Set-up rotation	[5, 5, 50] N m/rad, surface $[t_1, t_2, n_s]$	[10.01, 10.01, 10] N m s/rad
Hold translation	2000 N/m, isotropic base	50 N s/m
Hold rotation	50 N m/rad, isotropic base	8 N m s/rad
Manual guidance	no Cartesian stiffness; joint coordinates	0.5 N m s/rad joint damping

The active timing and trajectory values are listed in Table C.3.

Table C.3: Phase-transition and trajectory parameters.

Quantity	Configuration key	Value
Orient minimum time	approach_orient_min_time	1.0 s
Tool-axis switch tolerance	approach_orient_error_threshold	0.016 rad
Spin switch tolerance	approach_orient_spin_error_threshold	0.009 rad
Orient timeout / maximum rate	approach_orient_timeout, approach_orient_max_rate_deg	8.0 s, 20°/s
Descent speed / maximum distance	descend_speed, descend_max_distance	0.1 m/s, 0.640 m
Tool clearance, $h_{\text{clearance}}$	descend_surface_clearance	0.020 m
Set-up minimum time / timeout	setup_min_time, setup_timeout	2.0 s, 5.0 s
Set-up moment-change threshold	setup_moment_threshold	150 N m
Set-up push speed / endpoint	setup_push_speed, setup_push_end	0.080 m/s, 0.240 m in all surface-contact experiments
Pre-grind gate	pause_before_grind	enabled

C.3 Inertia-Scaled Damping and Centre of Compliance

Table C.4: Damping and compliance-centre parameters.

Quantity	Symbol / key	Value
Approach damping factor	ζ	1.9
Set-up damping factor	ζ	1.0
Task-inertia regularisation	added to $J(q)M(q)^{-1}J^\top(q)$	10^{-9}
Maximum damping coefficient	<code>auto_damping_max</code>	400
Lower bound from the configured matrices	<code>auto_damping_min_from_manual</code>	0; those matrices act as fallback only
Reported contact phases using inertia-scaled damping	—	approach and set-up
Pose-hold inertia-scaled damping	<code>hold_auto_damping</code>	0 in all twelve reported runs
Pose-hold translational damping	D_p	$50I_3$ N s/m
Pose-hold rotational damping	D_R	$8I_3$ N m s/rad
Coupled 6×6 law	<code>use_virtual_compliance_center</code>	set-up only, and only for displaced-centre conditions
Displacement definition frame	<code>compliance_center_in_tool_frame</code>	tool-fixed, except the surface-fixed conditions of Appendix D.3
Tool-fixed displacement	<code>compliance_center_offset_ee</code>	in end-effector axes
Surface-fixed displacement	<code>compliance_lever_surface</code>	in surface axes $[t_1, t_2, n_s]$

Both configured displacements define $r_c = p_c - p_{\text{TCP}}$ and differ only in the frame their components are held constant in. The damping calculation and the compliance-centre transformation are described in Section 3.2.4, and the tested displacement components are listed in Table 4.4 and Appendix D.

C.4 Null-Space and Operating Parameters

The null-space settings are explained in Section 4.3.2. The final two rows of Table C.5 are Franka collision/reflex thresholds configured through the libfranka collision-behaviour interface. Robot-side monitoring applied these thresholds throughout the reported runs.

Table C.5: Null-space and operating parameters.

Parameter	Symbol / key	Value
Null-space mode	<code>nullspace_mode</code>	3 (projected damping and singular-value conditioning)
Null-space damping	d_{null}	2.0 N m s/rad
Sigma torque magnitude	k_{σ}	1.0 N m
Probe distance	α_{probe}	0.08 rad
Sigma difference deadband	δ_{σ}	10^{-6}
Relative SVD cutoff	ρ	10^{-4}
Recorded-data sampling / capacity	<code>log_every_n_cycles</code> , rows	every callback (1), 120000
Run duration / stop	<code>experiment_duration</code>	0 s: indefinite until e+Enter
Gripper grasp	width, speed, force	0.066 m, 0.050 m/s, 60 N
Tool-mount rotational play	observed pad-mount clearance	approximately $\pm 2^\circ$ about y_{EE}
Manual-guidance damping	<code>manual_guidance_damping</code>	0.5 N m s/rad
Joint-side collision threshold	acceleration/nominal	140 N m on each axis
Cartesian collision threshold	acceleration/nominal	240 N or N m on each channel

Appendix D

Supporting Contact Checks and Results

This appendix reports the complete numerical values behind the main contact figures and four supporting checks. The supporting checks comprise 54 runs across 18 independently counted settings, with three repetitions per setting. The calibrated geometry, phase sequence, Cartesian gains, null-space settings, and evaluation quantities were those of Sections 4.2 and 4.3 unless a changed condition is stated below.

D.1 Numerical Values for Main Cases A–D

The main chapter retains the comparison figures and interpretation. The tables below provide the corresponding numerical values, each as the arithmetic mean and sample standard deviation over the three repetitions of that setting. In Table D.4 the reported coordinate is r_{c,t_2} for commands about t_1 and r_{c,t_1} for commands about t_2 , so the displacement selected for a positive offset appears at +40 mm in the upper table and at −40 mm in the lower one.

Table D.1: Case-A baseline response at the TCP, mean $\pm$ SD.

Commanded Initial Orientation	CoC Position	Measured Set-Up Rotation γ_{t_i} [°]
No orientation offset	TCP	-0.97 ± 0.02
+10° about t_1	TCP	$+7.57 \pm 0.01$
-10° about t_1	TCP	-10.83 ± 0.06
+10° about t_2	TCP	-1.63 ± 0.17
-10° about t_2	TCP	-1.25 ± 0.19

Table D.2: Case-B rotational-stiffness response, mean $\pm$ SD.

Commanded Initial Orientation	CoC Position	Varied Rotational Stiffness K_{R,t_i} [N m/rad]	Reference	Measured Set-Up Rotation γ_{t_i} [°]
+10° about t_1	TCP	5	Case-A reference	$+7.57 \pm 0.01$
		15	—	$+7.29 \pm 0.01$
		50	—	$+2.73 \pm 0.03$
+10° about t_2	TCP	5	Case-A reference	-1.63 ± 0.17
		15	—	-1.52 ± 0.31
		50	—	-0.48 ± 0.03

Table D.3: Case-C cross-axis translational-stiffness response, mean $\pm$ SD.

Commanded Initial Orientation	CoC Position	Varied K_{p,t_j}	K_p [N/m]	Reference	Measured Set-Up Rotation γ_{t_i} [°]
+10° about t_1	TCP	K_{p,t_2}	300	—	$+7.59 \pm 0.00$
		K_{p,t_2}	800	—	$+7.56 \pm 0.02$
		K_{p,t_2}	2000	Case-A reference	$+7.57 \pm 0.01$
+10° about t_2	TCP	K_{p,t_1}	300	—	-2.24 ± 0.14
		K_{p,t_1}	800	—	-1.78 ± 0.03
		K_{p,t_1}	2000	Case-A reference	-1.63 ± 0.17

Table D.4: Case-D tangential compliance-centre response, mean $\pm$ SD.

Commanded rotation about t_1 ; coordinate r_{c,t_2}							
Offset	-40	-20	-10	TCP	+10	+20	+40 mm
+10°	0.18 $\pm$ 0.07	0.99 $\pm$ 0.44	7.45 $\pm$ 0.02	7.57 $\pm$ 0.01	7.67 $\pm$ 0.01	7.73 $\pm$ 0.01	7.87 $\pm$ 0.01
-10°	-11.30 $\pm$ 0.11	-11.18 $\pm$ 0.09	-11.14 $\pm$ 0.07	-10.83 $\pm$ 0.06	-10.36 $\pm$ 0.21	-5.01 $\pm$ 0.07	-0.09 $\pm$ 0.01

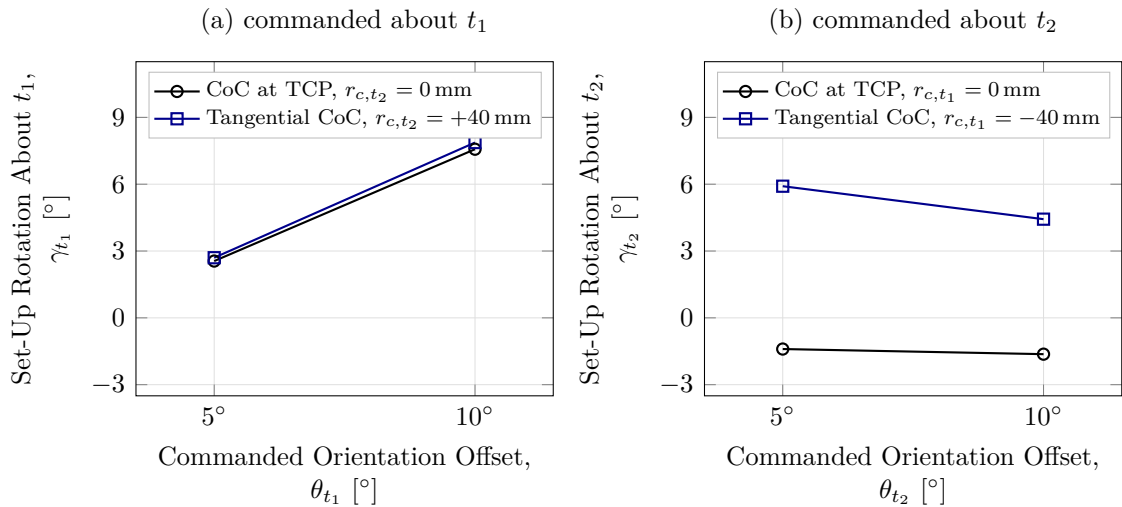
Commanded rotation about t_2 ; coordinate r_{c,t_1}							
Offset	-40	-20	-10	TCP	+10	+20	+40 mm
+10°	4.43 $\pm$ 0.07	-1.12 $\pm$ 0.09	-1.31 $\pm$ 0.13	-1.63 $\pm$ 0.17	-1.85 $\pm$ 0.36	-1.95 $\pm$ 0.47	-4.51 $\pm$ 2.36
-10°	4.68 $\pm$ 0.05	-1.28 $\pm$ 0.05	-1.29 $\pm$ 0.01	-1.25 $\pm$ 0.19	-1.29 $\pm$ 0.14	-1.51 $\pm$ 0.25	-10.51 $\pm$ 0.71

D.2 Orientation-Offset Magnitude

The 5° and 10° commanded offsets were compared with the CoC at the TCP and at the selected 40 mm tangential position. These four independently counted settings gave 12 runs.

Table D.5: Offset-magnitude response at two compliance-centre positions.

Rotation Direction	Varied Offset	TCP-Centred γ_{t_i} [°]	Selected 40 mm CoC γ_{t_i} [°]	Effect of the CoC [°]
About t_1	5°	+2.55	+2.70	+0.15
About t_1	10°	+7.57	+7.87	+0.30
About t_2	5°	-1.40	+5.91	+7.31
About t_2	10°	-1.63	+4.43	+6.06


Figure D.1: Set-up rotation by commanded orientation-offset magnitude.

The response did not scale proportionally with the commanded orientation-offset magnitude: about t_2 the selected displacement gave $+5.91^\circ$ at 5° and $+4.43^\circ$ at 10° , as Figure D.1 shows.

D.3 Compliance-Centre Definition Frame

This check compared the two displacement definitions of Equation (2.43) at the same nominal 40 mm tangential magnitude. The settings were the tool-fixed displacement without an orientation offset and the surface-fixed displacement with no offset or with a $+10^\circ$ offset about t_1 . These three settings gave 9 runs, and the shared tool-fixed $+10^\circ$ condition is counted with the $+40$ mm reference of Case D.

Table D.6: Definition-frame response at a common 40 mm magnitude.

Commanded Initial Orientation	Tool-Fixed Definition $\gamma_{t_1} [^\circ]$	Surface-Fixed Definition $\gamma_{t_1} [^\circ]$
None	-0.06 ± 0.01	-1.05 ± 0.02
$+10^\circ$ about t_1	$+7.87 \pm 0.01$	$+0.13 \pm 0.01$

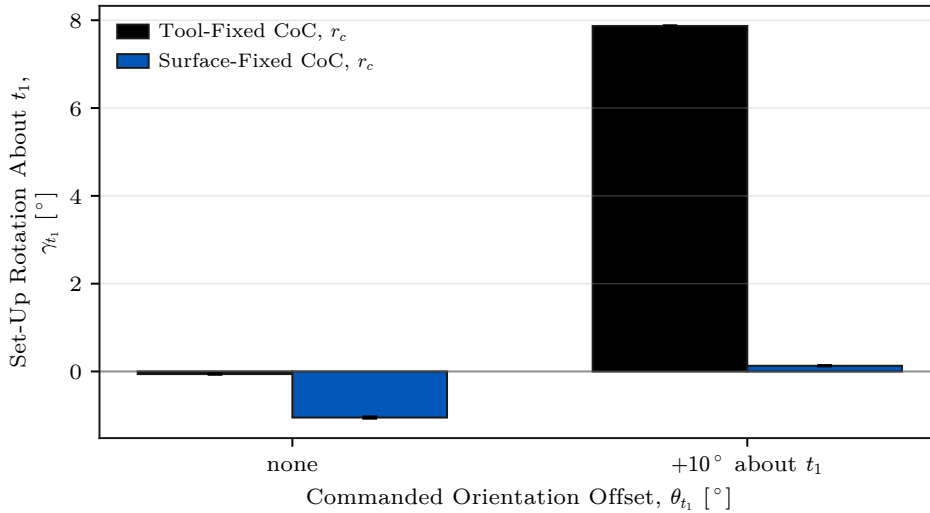


Figure D.2: Set-up rotation for tool- and surface-fixed displacement definitions.

The two definitions produced markedly different responses. For the $+10^\circ$ command about t_1 the tool-fixed definition gave $+7.87 \pm 0.01^\circ$ of set-up rotation, while the surface-fixed definition gave $+0.13 \pm 0.01^\circ$ and was classified as tilted by TCP height, as Table D.6

and Figure D.2 show. A non-zero CoC displacement is therefore not characterised by its magnitude alone.

D.4 Tool-Axis Displacement

This check separated the distance between the CoC and the TCP from the direction of the displacement. The centre was moved by -40 , -20 , -10 , $+10$, $+20$ and $+40$ mm along the tool axis at a fixed $+10^\circ$ command about t_1 . These six settings gave 18 runs, and the shared zero-displacement condition is counted with the Case-A reference.

Table D.7: Tool-axis displacement response at a $+10^\circ$ command about t_1 .

Measured Response	-40	-20	-10	0	$+10$	$+20$	$+40$ mm
γ_{t_1}	$+7.26$	$+7.49$	$+7.58$	$+7.57$	$+7.58$	$+7.55$	$+7.60$
Standard deviation	0.13	0.01	0.01	0.01	0.01	0.03	0.04

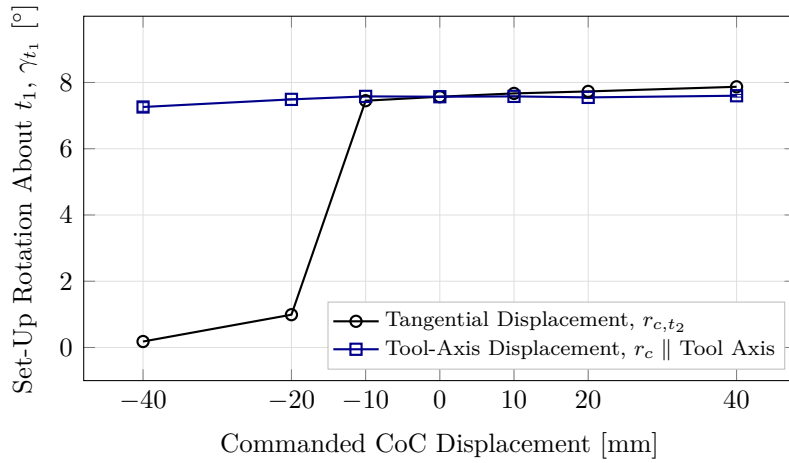


Figure D.3: Tangential and tool-axis centre displacement at the same command.

The signed set-up rotation changed from $+7.26$ to $+7.60^\circ$ between -40 and $+40$ mm, a span of 0.34° , against 7.73° for the purely tangential displacement of Case D over the same interval, as Figure D.3 shows. Every tool-axis position was classified as flat.

Only the tangential component of the CoC displacement contributes to the moment generated by the commanded normal force, by Equation (2.46). Let α_{axis} denote the acute inclination of the tool axis relative to the inward surface-normal direction $-n_s$, which is the direction the tool axis takes when the face is flat on the surface. The check used a

+10° commanded orientation offset about t_1 , so the tool axis was nominally inclined by 10° from $-n_s$ and a displacement along it carried a tangential magnitude

$$\|r_{c,t}\| = \|r_c\| \sin \alpha_{\text{axis}}. \quad (\text{D.1})$$

At the largest tested tool-axis displacement, $\|r_c\| = 40$ mm. Evaluating the projection at the nominal inclination, to two decimal places, gives

$$\|r_{c,t}\| = (40 \text{ mm}) \sin(10^\circ) = 6.95 \text{ mm}.$$

The 40 mm value is the total configured CoC displacement along the tool axis, and the 6.95 mm value only its projection onto the surface tangent plane. That effective tangential lever is considerably smaller than the tangential displacement of Case D, which is consistent with the weak variation measured here. Tangential force components, finite rotation, changing contact geometry and tool-mount compliance were not isolated by this check.

D.5 Intermediate Tangent-Plane Directions

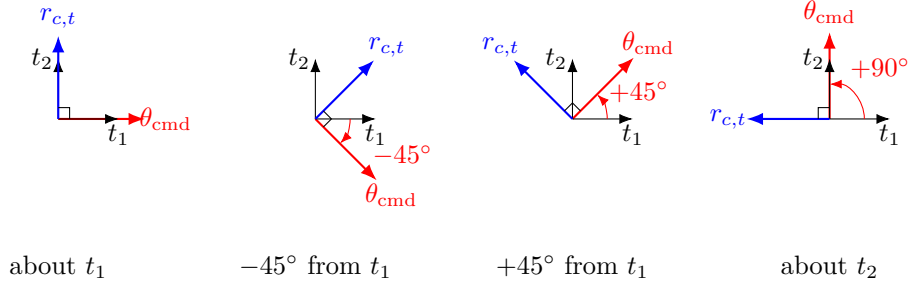
This check examined whether the direction rule of Case D also held between the principal surface tangents. For a commanded rotation with surface components $(\theta_{t_1}, \theta_{t_2})$, the selected tangential displacement was

$$r_{c,t} = \frac{\|r_{c,t}\|}{\sqrt{\theta_{t_1}^2 + \theta_{t_2}^2}} (\theta_{t_1} t_2 - \theta_{t_2} t_1), \quad (\text{D.2})$$

where $\|r_{c,t}\| = 40$ mm. The fixed comparison retained the vector selected for the t_1 command.

Table D.8: Commanded directions and the tangential displacements tested against them.

Commanded Rotation Direction	Displacement Setting	Commanded Rotation Components, $[\theta_{t_1}, \theta_{t_2}]$ [°]	CoC Components, $[r_{c,t_1}, r_{c,t_2}, r_{c,n}]$ [mm]
About t_1	selected	$[+10.0, 0.0]$	$[0, +40.0, 0]$
-45° from t_1	selected	$[+7.1, -7.1]$	$[+28.3, +28.3, 0]$
$+45^\circ$ from t_1	selected	$[+7.1, +7.1]$	$[-28.3, +28.3, 0]$
About t_2	selected	$[0.0, +10.0]$	$[-40.0, 0, 0]$
-45° from t_1	fixed at the t_1 selection	$[+7.1, -7.1]$	$[0, +40.0, 0]$
$+45^\circ$ from t_1	fixed at the t_1 selection	$[+7.1, +7.1]$	$[0, +40.0, 0]$
About t_2	fixed at the t_1 selection	$[0.0, +10.0]$	$[0, +40.0, 0]$


Figure D.4: Selected tangential displacement for each commanded rotation direction.

The principal-axis selected conditions are counted with Case D. The five supporting-check settings were both displacement definitions at each diagonal direction and the fixed displacement for the t_2 command, each repeated three times, giving 15 runs. Every non-zero setting was configured tool-fixed in end-effector coordinates, and the vectors in Table D.8 are the surface-frame components they realise in the flat target orientation.

Table D.9: Direction-rule comparison at $\|r_{c,t}\| = 40$ mm.

Commanded Rotation Direction [°]	Selected Displacement		Fixed at t_1 γ_{t_i} [°]
	$[r_{c,t_1}, r_{c,t_2}]$ [mm]	Selected γ_{t_i} [°]	
About t_1	$[0, +40.0]$	+7.87	+7.87
-45° from t_1	$[+28.3, +28.3]$	+4.46	+4.39
$+45^\circ$ from t_1	$[-28.3, +28.3]$	+4.85	+4.80
About t_2	$[-40.0, 0]$	+4.43	-1.61

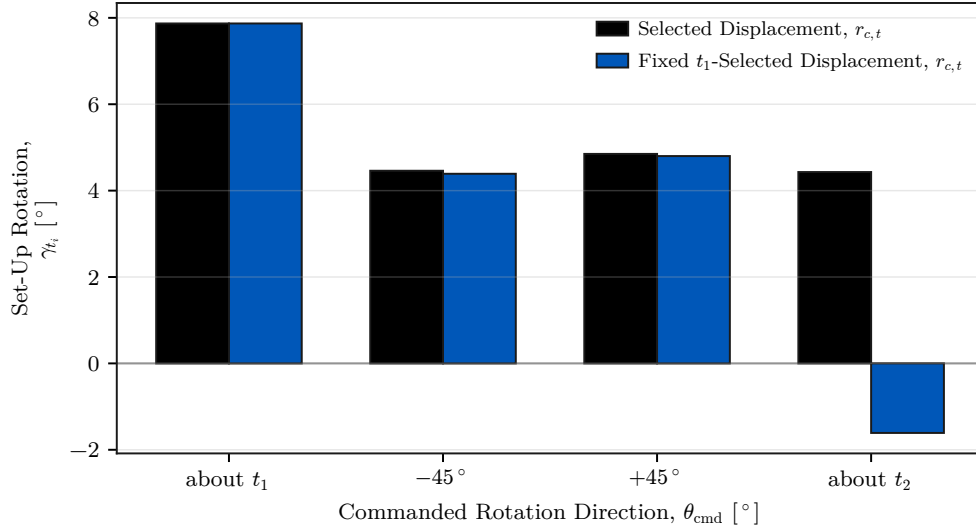


Figure D.5: Set-up rotation with the direction-selected and fixed displacements.

At the two diagonal directions the selected and fixed displacements differed by 0.07° and 0.05° , within the observed scatter, and the fixed displacement retained a substantial projection along the selected direction in both. The clearest separation occurred about t_2 , where Table D.9 and Figure D.5 give $+4.43^\circ$ for the selected displacement against -1.61° for the fixed one, a difference of 6.04° . These measurements support the direction rule of Section 2.7.2 over the four tested directions without establishing a single non-zero displacement that suits all of them.

D.6 Conditions Classified as Tilted by TCP Height

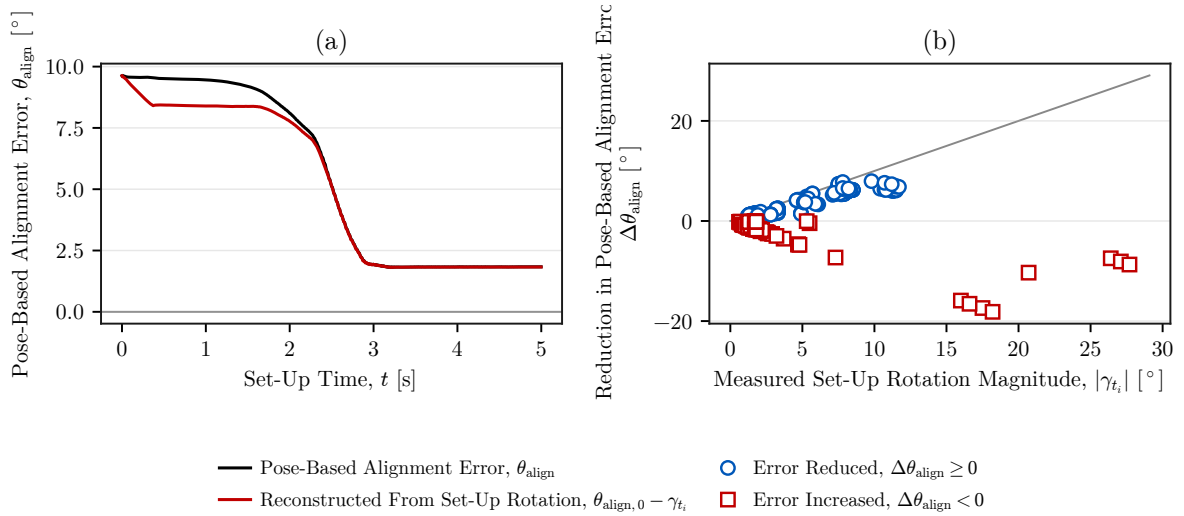
The TCP-height classification of Section 4.5.2 is retained here as a supporting geometric check and does not enter the response comparisons of Chapter 5. Of the 55 tested parameter settings, 49 satisfied it, including all five Case-A settings; Table D.10 lists the six that did not, with the excess TCP height above the flat-tool height.

Table D.10: Conditions classified as tilted by TCP height.

Condition	Excess height [mm]	γ_{t_1} [°]
Case B, 50 N m/rad about t_1	3.6	+2.73
Case D, -20 mm at +10° about t_1	5.3	+0.99
Case D, -40 mm at +10° about t_1	6.1	+0.18
Case D, +20 mm at -10° about t_1	5.2	-5.01
Case D, +40 mm at -10° about t_1	9.3	-0.09
Definition-frame check, surface-fixed at +10° about t_1	5.9	+0.13

The signed set-up rotation does not use the calibrated tool normal. It is obtained from the measured end-effector orientations and resolved in the surface frame, which is why it is the quantity compared between settings.

D.7 Pose-Based Alignment Consistency Check


Figure D.6: Pose-based alignment estimate against measured set-up rotation.

The main results use the signed set-up response γ_{t_i} . This appendix compares it with a secondary alignment estimate reconstructed from the calibrated tool normal,

$$\theta_{\text{align}}(t) = \cos^{-1}(-n_{\text{Tool},0}(t)^\top n_s), \quad n_{\text{Tool},0} = R_{\text{EE}} n_{\text{Tool},\text{EE}}. \quad (\text{D.3})$$

which is unsigned and depends on the assumed fixed relation between the tool and the

end effector. Its reduction over set-up is

$$\Delta\theta_{\text{align}} = \theta_{\text{align,before}} - \theta_{\text{align,after}}, \quad (\text{D.4})$$

taking $\theta_{\text{align,before}}$ at the start of set-up and $\theta_{\text{align,after}}$ from the settled set-up interval. It is used in this appendix alone.

Panel (a) plots the pose-based error for one representative trial against the value reconstructed from the measured rotation since the start of set-up, and panel (b) gives one point per evaluated trial, carrying the magnitude of the measured set-up rotation and the signed reduction of the pose-based error. Where the pose-based error decreased, its reduction followed the same scale as the measured rotation for most trials; conditions in which it increased appear below zero.

Their agreement is an internal consistency check rather than an independent measurement of the physical tool orientation, because the pose-based quantity depends on the assumed tool-to-gripper relation while γ_{t_i} is obtained directly from the measured end-effector orientations. The approximately $\pm 2^\circ$ of mounting play was measured unloaded and was not characterised as an uncertainty bound under contact load.